

Bibliography smoke test

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References

- Marwa Abdulhai, Isadora White, Charlie Snell, Charles Sun, Joey Hong, Yuexiang Zhai, Kelvin Xu, and Sergey Levine. 2023. LMRL Gym: Benchmarks for Multi-Turn Reinforcement Learning with Language Models. (2023). [arXiv:2311.18232](https://arxiv.org/abs/2311.18232)
- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, and Shyamal Anadkat. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).
- Eshaan Agarwal, Joykirat Singh, Vivek Dani, Raghav Magazine, Tanuja Ganu, and Akshay Nambi. 2024. PromptWizard: Task-Aware Agent-driven Prompt Optimization Framework. *arXiv preprint arXiv:2405.18369* (2024).
- Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziem, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, and Meng Jiang. 2025. Gega: Reflective prompt evolution can outperform reinforcement learning. *arXiv preprint arXiv:2507.19457* (2025).
- Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, Keerthana Gopalakrishnan, and Karol Hausman. 2022. Do as i can, not as i say: Grounding language in robotic affordances. *arXiv preprint arXiv:2204.01691* (2022).
- Bakhtawar Ahtisham, Kirk Vanacore, Jinsook Lee, Zhuqian Zhou, Doug Pietrzak, and Rene F Kizilcec. 2026. Ai annotation orchestration: Evaluating llm verifiers to improve the quality of llm annotations in learning analytics. In *Proceedings of the LAK26: 16th International Learning Analytics and Knowledge Conference*. 447–456.
- Afra Feyza Akyurek, Ekin Akyurek, Aman Madaan, Ashwin Kalyan, Peter Clark, Derry Wijaya, and Niket Tandon. 2023. RL4F: Generating Natural Language Feedback with Reinforcement Learning for Repairing Model Outputs. In *Annual Meeting of the Association for Computational Linguistics*.
- Lisa Alazraki, Tan Yi-Chern, Jon Ander Campos, Maximilian Mozes, Marek Rei, and Max Bartolo. 2025. Reverse Engineering Human Preferences with Reinforcement Learning. [arXiv:2505.15795](https://arxiv.org/abs/2505.15795)
- Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. 2016. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565* (2016).
- Markus Anderljung, Joslyn Barnhart, Anton Korinek, Jade Leung, Cullen O’Keefe, Jess Whittlestone, Shahar Avin, Miles Brundage, Justin Bullock, and Duncan Cass-Beggs. 2023. Frontier AI regulation: Managing emerging risks to public safety. *arXiv preprint arXiv:2307.03718* (2023).
- Jacob Andreas, Dan Klein, and Sergey Levine. 2017. Modular Multitask Reinforcement Learning with Policy Sketches. In *International Conference on Machine Learning*. 166–175. [arXiv:1611.01796](https://arxiv.org/abs/1611.01796)
- Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, OpenAI Pieter Abbeel, and Wojciech Zaremba. 2017. Hindsight experience replay. *Advances in neural information processing systems* 30 (2017). [arXiv:1707.01495](https://arxiv.org/abs/1707.01495)
- Anonymous. 2025. Self-Improvement of Large Language Models: A Technical Overview and Future Outlook. *arXiv preprint* (2025).
- Samuel Arnesen, David Rein, and Julian Michael. 2024. Training Language Models to Win Debates with Self-Play Improves Judge Accuracy. (2024). [arXiv:2409.16636](https://arxiv.org/abs/2409.16636)
- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. *arXiv preprint arXiv:2310.11511* (2023).
- Mohammad Gheshlaghi Azar, Zhaohan Daniel Guo, Bilal Piot, Remi Munos, Mark Rowland, Michal Valko, and Daniele Calandriello. 2024. A general theoretical paradigm to understand learning from human preferences. In *International Conference on Artificial Intelligence and Statistics*. PMLR, 4447–4455.
- Juyang Bai and Laixi Shi. 2026. MAS-PromptBench: When Does Prompt Optimization Improve Multi-Agent LLM Systems? (2026). [arXiv:2606.23664](https://arxiv.org/abs/2606.23664)
- Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, and Tom Henighan. 2022a. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862* (2022).
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Christopher Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt, Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Noemi Mercado, Nova DasSarma, Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna Kravec, Sheer El Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly, Tom Henighan, Tristan Hume, Samuel R Bowman, Zac Hatfield-Dodds, Ben Mann, Dario

Author’s Contact Information: None, None, Minneapolis, USA.

2026. Manuscript submitted to ACM

Manuscript submitted to ACM

- Amodei, Nicholas Joseph, Sam McCandlish, Tom Brown, and Jared Kaplan. 2022b. Constitutional AI: Harmlessness from AI Feedback. *arXiv preprint arXiv:2212.08073* (2022).
- Jayachandru Bandlamudi, Ritwik Chaudhuri, Neelamadhav Gantayat, Sambit Ghosh, Kushal Mukherjee, Perna Agarwal, Renuka Sindhgatta, and Sameep Mehta. 2025. A framework for testing and adapting rest apis as llm tools. *arXiv preprint arXiv:2504.15546* (2025).
- Tim Baumgärtner, Yang Gao, Dana Alon, and Donald Metzler. 2024. Best-of-Venom: Attacking RLHF by Injecting Poisoned Preference Data. *arXiv:2404.05530*
- Mohammad Beigi, Ying Shen, Parshin Shojaei, Qifan Wang, Zichao Wang, Chandan Reddy, Ming Jin, and Lifu Huang. 2025. Sycophancy Mitigation Through Reinforcement Learning with Uncertainty-Aware Adaptive Reasoning Trajectories. *arXiv:2509.16742*
- Julia Belikova, Rauf Parchiev, Evgeny Egorov, Grigori Davydenko, Gleb Gusev, Andrey Savchenko, and Maksim Makarenko. 2026. Managing Procedural Memory in LLM Agents: Control, Adaptation, and Evaluation. (2026). *arXiv:2606.23127*
- Suneel Belkale, Tianli Ding, Ted Xiao, Pierre Sermanet, Quon Vuong, Jonathan Tompson, Yevgen Chebotar, Debidatta Dwibedi, and Dorsa Sadigh. 2024. RT-H: Action Hierarchies Using Language. (2024). *arXiv:2403.01823*
- Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, and Piotr Nyczyk. 2024. Graph of thoughts: Solving elaborate problems with large language models. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 38. 17682–17690.
- Luca Beurer-Kellner, Beat Buesser, Ana-Maria Crețu, Edoardo DeBenedetti, Daniel Dobos, Daniel Fabian, Marc Fischer, David Froelicher, Kathrin Grosse, and Daniel Naef. 2025. Design patterns for securing llm agents against prompt injections. *arXiv preprint arXiv:2506.08837* (2025).
- Vineet Bhat, Ali Umut Kaypak, Prashanth Krishnamurthy, Ramesh Karri, and Farshad Khorrami. 2024. Grounding llms for robot task planning using closed-loop state feedback. *arXiv preprint arXiv:2402.08546* (2024).
- Johan Björck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi Fan, Yu Fang, Dieter Fox, Fengyuan Hu, and Spencer Huang. 2025. Gr00t n1: An open foundation model for generalist humanoid robots. *arXiv preprint arXiv:2503.14734* (2025).
- Beatriz Borges, Negar Foroutan, Deniz Bayazit, Anna Sotnikova, Syrielle Montariol, Tanya Nazaretzky, Mohammadreza Banaei, Alireza Sakhaeirad, Philippe Servant, Seyed Parsa Neshaei, and Jibril Frej. 2024. Let Me Teach You: Pedagogical Foundations of Feedback for Language Models. In *Conference on Empirical Methods in Natural Language Processing*.
- Avinandan Bose, Zhihan Xiong, Aadirupa Saha, Simon Shao-lei Du, and Maryam Fazel. 2024. Hybrid preference optimization for alignment: Provably faster convergence rates by combining offline preferences with online exploration. *arXiv preprint arXiv:2412.10616* (2024).
- Andres Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D White, and Philippe Schwaller. 2024. Augmenting large language models with chemistry tools. *Nature machine intelligence* 6, 5 (2024), 525–535.
- Satchuthananthavale RK Branavan, Harr Chen, Luke Zettlemoyer, and Regina Barzilay. 2009. Reinforcement learning for mapping instructions to actions. In *Proceedings of the Joint Conference of the 47th Annual Meeting of the ACL and the 4th International Joint Conference on Natural Language Processing of the AFNLP*. 82–90.
- Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, D Driess, A Dubey, and C Finn. 2023. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv. arXiv preprint arXiv:2307.15818* (2023).
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, and Amanda Askell. 2020. Language models are few-shot learners. *Advances in neural information processing systems* 33 (2020), 1877–1901.
- Alexander Bukharin, Haifeng Qian, Shengyang Sun, Adithya Renduchintala, Soumye Singhal, Zhilin Wang, Oleksii Kuchaiev, Olivier Delalleau, and Duo Zhao. 2025. Adversarial Training of Reward Models. *arXiv:2504.06141*
- Yuzheng Cai, Siqi Cai, Yuchen Shi, Zihan Xu, Lichao Chen, Yulei Qin, Xiaoyu Tan, Gang Li, Zongyi Li, Haojia Lin, Yong Mao, Ke Li, and Xing Sun. 2025a. Training-Free Group Relative Policy Optimization. (2025). *arXiv:2510.08191*
- Zhicheng Cai, Xinyuan Guo, Yu Pei, Jiangtao Feng, Jinsong Su, Jiangjie Chen, Ya-Qin Zhang, Wei-Ying Ma, and Mingxuan Wang. 2025b. FLEX: Continuous Agent Evolution via Forward Learning from Experience. *arXiv preprint arXiv:2511.06449* (2025). *arXiv:2511.06449*
- Zouying Cao, Jiaji Deng, Li Yu, Weikang Zhou, Zhaoyang Liu, Bolin Ding, and Hai Zhao. 2025. Remember Me, Refine Me: A Dynamic Procedural Memory Framework for Experience-Driven Agent Evolution. (2025). *arXiv:2512.10696*
- Stephen Casper, Xander Davies, Claudia Shi, Thomas Krendl Gilbert, Jérémy Scheurer, Javier Rando, Rachel Freedman, Tomasz Korbak, David Lindner, and Pedro Freire. 2023. Open problems and fundamental limitations of reinforcement learning from human feedback. *arXiv preprint arXiv:2307.15217* (2023).
- Yekun Chai, Haoran Sun, Huang Fang, Shuohuan Wang, Yu Sun, and Hua Wu. 2024. MA-RLHF: Reinforcement Learning from Human Feedback with Macro Actions. (2024). *arXiv:2410.02743*
- Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. 2023. ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate. *arXiv:2308.07201* [cs.CL] <https://arxiv.org/abs/2308.07201>
- Devendra Singh Chaplot, Kanthashree Mysore Sathyendra, Rama Kumar Pasumarthi, Dheeraj Rajagopal, and Ruslan Salakhutdinov. 2018. Gated-attention architectures for task-oriented language grounding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 32.
- Angelica Chen, Jérémy Scheurer, Jon Ander Campos, Tomasz Korbak, Jun Shern Chan, Samuel R Bowman, Kyunghyun Cho, and Ethan Perez. 2024b. Learning from Natural Language Feedback. *Transactions on Machine Learning Research* (2024).
- Bei Chen, Fengji Zhang, Anh Nguyen, Daoguang Zan, Zeqi Lin, Jian-Guang Lou, and Weizhu Chen. 2022. CodeT: Code Generation with Generated Tests. *arXiv preprint arXiv:2207.10397* (2022).
- Manuscript submitted to ACM

- Hongyu Chen and Seraphina Goldfarb-Tarrant. 2025. Safer or Luckier? LLMs as Safety Evaluators Are Not Robust to Artifacts. [arXiv:2503.09347](#)
- Hao Chen, Ziyu Han, Yukun Yan, Qingfu Zhu, Maosong Sun, and Wanxiang Che. 2026b. From Holistic Evaluation to Structured Criteria: Rubrics Across the Evolving LLM Landscape. (2026). [arXiv:2606.08625](#)
- Jiaqi Chen, Bang Zhang, Ruotian Ma, Peisong Wang, Xiaodan Liang, Zhaopeng Tu, Xiaolong Li, and Kwan-Yee K. Wong. 2025c. SPC: Evolving Self-Play Critic via Adversarial Games for LLM Reasoning. (2025). [arXiv:2504.19162](#)
- Minghui Chen, Wenlong Deng, James Zou, Han Yu, and Xiaoxiao Li. 2026a. Textual Equilibrium Propagation for Deep Compound AI Systems. (2026). [arXiv:2601.21064](#)
- Sizhe Chen, Arman Zharmagambetov, Saeed Mahloujifar, Kamalika Chaudhuri, David Wagner, and Chuan Guo. 2025d. SecAlign: Defending Against Prompt Injection with Preference Optimization. [arXiv:2410.05451 \[cs.CR\]](#) [doi:10.1145/3719027.3744836](#)
- Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. 2020. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*. PmLR, 1597–1607. [arXiv:2002.05709](#)
- Xiushi Chen, Gaotang Li, Ziqi Wang, Bowen Jin, Cheng Qian, Yu Wang, Hongru Wang, Yu Zhang, Denghui Zhang, Tong Zhang, Hanghang Tong, and Heng Ji. 2025a. RM-R1: Reward Modeling as Reasoning. (2025). [arXiv:2505.02387](#)
- Xinyun Chen, Maxwell Lin, Nathanael Schärli, and Denny Zhou. 2024a. Teaching large language models to self-debug. In *International Conference on Learning Representations*, Vol. 2024. 8746–8825.
- Xiaoyin Chen, Canwen Xu, Yite Wang, Boyi Liu, Zhewei Yao, and Yuxiong He. 2026c. Learning to Self-Evolve. (2026). [arXiv:2603.18620](#)
- Yixing Chen, Yiding Wang, Siqi Zhu, Haofei Yu, Tao Feng, Muhan Zhang, Mostofa Patwary, and Jiaxuan You. 2025b. Multi-Agent Evolve: LLM Self-Improve through Co-evolution. (2025). [arXiv:2510.23595](#)
- Ching-An Cheng, Andrey Kolobov, Dipendra Misra, Allen Nie, and Adith Swaminathan. 2023. LLF-Bench: Benchmark for Interactive Learning from Language Feedback. (2023). [arXiv:2312.06853](#)
- Ching-An Cheng, Allen Nie, and Adith Swaminathan. 2024b. Trace is the Next AutoDiff: Generative Optimization with Rich Feedback, Execution Traces, and LLMs. (2024). [arXiv:2406.16218](#)
- Myra Cheng, Sunny Yu, Cinoo Lee, Pranav Khadpe, Lujain Ibrahim, and Dan Jurafsky. 2025. ELEPHANT: Measuring and understanding social sycophancy in LLMs. [arXiv:2505.13995](#)
- Pengyu Cheng, Tianhao Hu, Han Xu, Zhisong Zhang, Zheng Yuan, Yong Dai, Lei Han, Nan Du, and Xiaolong Li. 2024a. Self-playing Adversarial Language Game Enhances LLM Reasoning. (2024). [arXiv:2404.10642](#)
- Maxime Chevalier-Boisvert, Dzmitry Bahdanau, Salem Lahlou, Lucas Willems, Chitwan Saharia, Thien Huu Nguyen, and Yoshua Bengio. 2018. Babyai: A platform to study the sample efficiency of grounded language learning. *arXiv preprint arXiv:1810.08272* (2018).
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory. (2025). [arXiv:2504.19413](#)
- Hyeong Kyu Choi, Xiaojin Zhu, and Yixuan Li. 2025. Debate or Vote: Which Yields Better Decisions in Multi-Agent Large Language Models? *Advances in Neural Information Processing Systems* (2025). [arXiv:2508.17536](#)
- Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep reinforcement learning from human preferences. *Advances in Neural Information Processing Systems* 30 (2017). [arXiv:1706.03741](#)
- Vanya Cohen, Geraud Nangue Tasse, Nakul Gopalan, Steven James, Matthew Gombolay, Ray Mooney, and Benjamin Rosman. 2025. Compositional Instruction Following with Language Models and Reinforcement Learning. *arXiv preprint arXiv:2501.12539* (2025).
- Marc-Alexandre Côté, Akos Kádár, Xingdi Yuan, Ben Kybartas, Tavian Barnes, Emery Fine, James Moore, Matthew Hausknecht, Layla El Asri, and Mahmoud Adada. 2018. Textworld: A learning environment for text-based games. In *Workshop on Computer Games*. Springer, 41–75.
- Yuchen Cui, Siddharth Karamcheti, Raj Pallei, Nidhya Shivakumar, Percy Liang, and Dorsa Sadigh. 2023. “No, to the Right” – Online Language Corrections for Robotic Manipulation via Shared Autonomy. (2023). [arXiv:2301.02555](#)
- Jeff Da, Clinton Wang, Xiang Deng, Yuntao Ma, Nikhil Barhate, and Sean Hendryx. 2025. Agent-RLVR: Training Software Engineering Agents via Guidance and Environment Rewards. (2025). [arXiv:2506.11425](#)
- Yinpei Dai, Jayjun Lee, Nima Fazeli, and Joyce Chai. 2024. RACER: Rich Language-Guided Failure Recovery Policies for Imitation Learning. (2024). [arXiv:2409.14674](#)
- Parth Darshan and Abhishek Divekar. 2026. When Gradients Collide: Failure Modes of Multi-Objective Prompt Optimization for LLM Judges. (2026). [arXiv:2605.26046](#)
- Sarkar Snigdha Sarathi Das, Ryo Kamoi, Bo Pang, Yusen Zhang, Caiming Xiong, and Rui Zhang. 2024. GReaTer: Gradients over Reasoning Makes Smaller Language Models Strong Prompt Optimizers. (2024). [arXiv:2412.09722](#)
- Edoardo Debenedetti, Jie Zhang, Mislav Balunovic, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. 2024. Agentdojo: A dynamic environment to evaluate prompt injection attacks and defenses for llm agents. *Advances in Neural Information Processing Systems* 37 (2024), 82895–82920.
- Jonas Degraeve, Federico Felici, Jonas Buchli, Michael Neunert, Brendan Tracey, Francesco Carpanese, Timo Ewalds, Roland Hafner, Abbas Abdolmaleki, and Diego de Las Casas. 2022. Magnetic control of tokamak plasmas through deep reinforcement learning. *Nature* 602, 7897 (2022), 414–419.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. [arXiv:1810.04805 \[cs.CL\]](#) <https://arxiv.org/abs/1810.04805>
- Shen Dong, Shaochen Xu, Pengfei He, Yige Li, Jiliang Tang, Tianming Liu, Hui Liu, and Zhen Xiang. 2026b. Memory injection attacks on LLM agents via query-only interaction. *Advances in Neural Information Processing Systems* 38 (2026), 46697–46731.

- Zhichen Dong, Yang Li, Yuhao Sun, Weixun Wang, Yijia Luo, Zinian Peng, Taiheng Ye, Chao Yang, Wenbo Su, Yu Cheng, Bo Zheng, and Junchi Yan. 2026a. How Does Reasoning Flow? Tracing Attention-Induced Information Flow for Targeted RL in LLMs. (2026). [arXiv:2606.10646](#)
- Danny Driess, Fei Xia, Mehdi SM Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, and Tianhe Yu. 2023. Palm-e: An embodied multimodal language model. *arXiv preprint arXiv:2303.03378* (2023).
- Lingxiao Du, Fanqing Meng, Zongkai Liu, Zhixiang Zhou, Ping Luo, Qiaosheng Zhang, and Wenqi Shao. 2025. MM-PRM: Enhancing Multimodal Mathematical Reasoning with Scalable Step-Level Supervision. (2025). [arXiv:2505.13427](#)
- Pengfei Du. 2026. Memory for autonomous llm agents: Mechanisms, evaluation, and emerging frontiers. *arXiv preprint arXiv:2603.07670* (2026).
- Shangheng Du, Jiabao Zhao, Jinxin Shi, Zhentao Xie, Xin Jiang, Yanhong Bai, and Liang He. 2026. A Survey on the Optimization of Large Language Model-based Agents. *Comput. Surveys* 58, 9 (2026). [arXiv:2503.12434](#)
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B Tenenbaum, and Igor Mordatch. 2024. Improving Factuality and Reasoning in Language Models through Multiagent Debate. In *International Conference on Machine Learning*.
- Jiafei Duan, Wilbert Pumacay, Nishanth Kumar, Yi Ru Wang, Shulin Tian, Wentao Yuan, Ranjay Krishna, Dieter Fox, Ajay Mandlekar, and Yijie Guo. 2024. AHA: A Vision-Language-Model for Detecting and Reasoning Over Failures in Robotic Manipulation. (2024). [arXiv:2410.00371](#)
- Mateusz Dziemian, Maxwell Lin, Xiaohan Fu, Micha Nowak, Nick Winter, Eliot Jones, Andy Zou, Lama Ahmad, Kamalika Chaudhuri, Sahana Chennabasappa, Xander Davies, Lauren Deason, Benjamin L. Edelman, Tanner Emek, Ivan Evtimov, Jim Gust, Maia Hamin, Kat He, Klaudia Krawiecka, Riccardo Patana, Neil Perry, Troy Peterson, Xiangyu Qi, Javier Rando, Zifan Wang, Zihan Wang, Spencer Whitman, Eric Winsor, Arman Zharmagambetov, Matt Fredrikson, and Zico Kolter. 2026. How Vulnerable Are AI Agents to Indirect Prompt Injections? Insights from a Large-Scale Public Competition. [arXiv:2603.15714](#)
- Andrew Estornell and Yang Liu. 2024. Multi-llm debate: Framework, principals, and interventions. *Advances in Neural Information Processing Systems* 37 (2024), 28938–28964.
- Kawin Ethayarajh, Winnie Xu, Niklas Muennighoff, Dan Jurafsky, and Douwe Kiela. 2024. Kto: Model alignment as prospect theoretic optimization. *arXiv preprint arXiv:2402.01306* (2024).
- Ivan Evtimov, Arman Zharmagambetov, Aaron Grattafiori, Chuan Guo, and Kamalika Chaudhuri. 2025. WASP: Benchmarking Web Agent Security Against Prompt Injection Attacks. [arXiv:2504.18575](#)
- Jinyuan Fang, Yanwen Peng, Xi Zhang, Yingxu Wang, Xinhao Yi, Guibin Zhang, Yi Xu, Bin Wu, Siwei Liu, Zihao Li, Zhaochun Ren, Nikos Aletras, Xi Wang, Han Zhou, and Zaiqiao Meng. 2025b. A Comprehensive Survey of Self-Evolving AI Agents: A New Paradigm Bridging Foundation Models and Lifelong Agentic Systems. (2025). [arXiv:2508.07407](#)
- Runnan Fang, Yuan Liang, Xiaobin Wang, Jialong Wu, Shuofei Qiao, Pengjun Xie, Fei Huang, Huajun Chen, and Ningyu Zhang. 2025a. Memp: Exploring Agent Procedural Memory. (2025). [arXiv:2508.06433](#)
- Aaron Fanous, Jacob Goldberg, Ank A. Agarwal, Joanna Lin, Anson Zhou, Roxana Daneshjou, and Sanmi Koyejo. 2025. SycEval: Evaluating LLM Sycophancy. [arXiv:2502.08177](#)
- Jiazhan Feng, Shijue Huang, Xingwei Qu, Ge Zhang, Yujia Qin, Baoquan Zhong, Chengquan Jiang, Jinxin Chi, and Wanjuan Zhong. 2025a. Retool: Reinforcement learning for strategic tool use in llms. *arXiv preprint arXiv:2504.11536* (2025).
- Xidong Feng, Bo Liu, Yan Song, Haotian Fu, Ziyu Wan, Girish A. Koushik, Zhiyuan Hu, Mengyue Yang, Ying Wen, and Jun Wang. 2024. Natural Language Reinforcement Learning. (2024). [arXiv:2411.14251](#)
- Yuanning Feng, Sinan Wang, Zhengxiang Cheng, Yao Wan, and Dongping Chen. 2025b. Are We on the Right Way to Assessing LLM-as-a-Judge? (2025). [arXiv:2512.16041](#)
- Patrick Fernandes, Aman Madaan, Emmy Liu, António Farinhas, Pedro Henrique Martins, Amanda Bertsch, José G. C. de Souza, Shuyan Zhou, Tongshuang Wu, Graham Neubig, and André F. T. Martins. 2023. Bridging the Gap: A Survey on Integrating (Human) Feedback for Natural Language Generation. (2023). [arXiv:2305.00955](#)
- Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. 2023. Promptbreeder: Self-Referential Self-Improvement Via Prompt Evolution. *arXiv preprint arXiv:2309.16797* (2023).
- Thomas Palmeira Ferraz, Kartik Mehta, Yu-Hsiang Lin, Haw-Shiuan Chang, Shereen Oraby, Sijia Liu, Vivek Subramanian, Tagyoung Chung, Mohit Bansal, and Nanyun Peng. 2024. LLM Self-Correction with DeCRIM: Decompose, Critique, and Refine for Enhanced Following of Instructions with Multiple Constraints. *arXiv preprint arXiv:2410.06458* (2024).
- Benjamin Feuer, Chiung-Yi Tseng, Astitwa Sarthak Lathe, Oussama Elachqar, and John P Dickerson. 2025. When Judgment Becomes Noise: How Design Failures in LLM Judge Benchmarks Silently Undermine Validity. (2025). [arXiv:2509.20293](#)
- Emily First, Markus N Rabe, Talia Ringer, and Yuriy Brun. 2023. Baldur: Whole-Proof Generation and Repair with Large Language Models. *arXiv preprint arXiv:2303.04910* (2023).
- Evan Frick, Tianle Li, Connor Chen, Wei-Lin Chiang, Anastasios N. Angelopoulos, Jiantao Jiao, Banghua Zhu, Joseph E. Gonzalez, and Ion Stoica. 2024. How to Evaluate Reward Models for RLHF. (2024). [arXiv:2410.14872](#)
- Huan-ang Gao, Jiayi Geng, Wenyue Hua, Mengkang Hu, Xinzhe Juan, Hongzhang Liu, Shilong Liu, Jiahao Qiu, Xuan Qi, Yiran Wu, Hongru Wang, Han Xiao, Yuhang Zhou, Shaokun Zhang, Jiayi Zhang, Jinyu Xiang, Yixiong Fang, Qiwen Zhao, Dongrui Liu, Qihan Ren, Cheng Qian, Zhenhailong Wang, Minda Hu, Huazheng Wang, Qingyun Wu, Heng Ji, and Mengdi Wang. 2025. A Survey of Self-Evolving Agents: What, When, How, and Where to Evolve on the Path to Artificial Super Intelligence. (2025). [arXiv:2507.21046](#)
- Manuscript submitted to ACM

- Luyu Gao, Zhuyun Dai, Panupong Pasupat, Anthony Chen, Arun Tejasvi Chaganty, Yicheng Fan, Vincent Zhao, Ni Lao, Hongrae Lee, Da-Cheng Juan, and Kelvin Guu. 2023. RARR: Researching and Revising What Language Models Say, Using Language Models. *arXiv preprint arXiv:2210.08726* (2023).
- Jonas Gehring, Kunhao Zheng, Jade Copet, Vegard Mella, Quentin Carbonneaux, Taco Cohen, and Gabriel Synnaeve. 2024. Rlef: Grounding code llms in execution feedback with reinforcement learning. *arXiv preprint arXiv:2410.02089* (2024).
- Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. 2024a. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing. In *International Conference on Learning Representations*. arXiv:2305.11738
- Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Minlie Huang, Nan Duan, and Weizhu Chen. 2024b. ToRA: A Tool-Integrated Reasoning Agent for Mathematical Problem Solving. arXiv:2309.17452 [cs.CL] <https://arxiv.org/abs/2309.17452>
- Joel Greenspoon. 1955. The reinforcing effect of two spoken sounds on the frequency of two responses. *The American Journal of Psychology* 68, 3 (1955), 409–416.
- Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. 2023. Not what you’ve signed up for: Compromising real-world llm-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM workshop on artificial intelligence and security*. 79–90.
- Jiawei Gu, Xuhui Jiang, Zhichao Shi, Hexiang Tan, Xuehao Zhai, Chengjin Xu, Wei Li, Yinghan Shen, Shengjie Ma, Honghao Liu, Saizhuo Wang, Kun Zhang, Yuanzhuo Wang, Wen Gao, Lionel Ni, and Jian Guo. 2024. A Survey on LLM-as-a-Judge. (2024). arXiv:2411.15594
- Xin Guan, Xiaomeng Hu, Shen Huang, Zhenyi Wang, Bo Zhang, Zijian Li, Pengjun Xie, Bo Liu, and Jiuxin Cao. 2026. EvoRubric: Self-Evolving Rubric-Driven RL for Open-Ended Generation. (2026). arXiv:2605.29847
- Caglar Gulcehre, Tom Le Paine, Srivatsan Srinivasan, Ksenia Konyushkova, Lotte Weerts, Abhishek Sharma, Aditya Siddhant, Alex Ahern, Miaosen Wang, and Chenjie Gu. 2023. Reinforced Self-Training (ReST) for Language Modeling. *arXiv preprint arXiv:2308.08998* (2023).
- Anisha Gunjal, Anthony Wang, Elaine Lau, Vaskar Nath, Yunzhong He, Bing Liu, and Sean Hendryx. 2025. Rubrics as Rewards: Reinforcement Learning Beyond Verifiable Domains. (2025). arXiv:2507.17746
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, and Xiao Bi. 2025b. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948* (2025).
- Jiaxin Guo, Zewen Chi, Li Dong, Qingxiu Dong, Xun Wu, Shaohan Huang, and Furu Wei. 2025a. Reward Reasoning Model. (2025). arXiv:2505.14674
- Qingyan Guo, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. 2024. Connecting Large Language Models with Evolutionary Algorithms Yields Powerful Prompt Optimizers. In *International Conference on Learning Representations*. arXiv:2309.08532
- Siyuan Guo, Yali Du, Hechang Chen, Yi Chang, and Jun Wang. 2026. CASCADE: Case-Based Continual Adaptation for Large Language Models During Deployment. (2026). arXiv:2605.06702
- Priyanshu Gupta, Shashank Kirtania, Ananya Singha, Sumit Gulwani, Arjun Radhakrishna, Sherry Shi, and Gustavo Soares. 2024. MetaReflection: Learning Instructions for Language Agents using Past Reflections. (2024). arXiv:2405.13009
- Dongyoon Hahn, Dylan Hadfield-Menell, and Kimin Lee. 2026. Alignment Tampering: How Reinforcement Learning from Human Feedback Is Exploited to Optimize Misaligned Biases. arXiv:2605.27355
- Hojae Han, Seung-won Hwang, Rajhans Samdani, and Yuxiong He. 2025. ConvCodeWorld: Benchmarking Conversational Code Generation in Reproducible Feedback Environments. (2025). arXiv:2502.19852
- Shibo Hao, Yi Gu, Haodi Ma, Joshua Hong, Zhen Wang, Daisy Wang, and Zhiting Hu. 2023. Reasoning with language model is planning with world model. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 8154–8173.
- Ali Hatamizadeh, Shrimai Prabhunoye, Igor Gitman, Ximing Lu, Seungju Han, Wei Ping, Yejin Choi, and Jan Kautz. 2026. iGRPO: Self-feedback-driven LLM reasoning. *arXiv preprint arXiv:2602.09000* (2026).
- Matthew Hausknecht, Prithviraj Ammanabrolu, Marc-Alexandre Côté, and Xingdi Yuan. 2020. Interactive Fiction Games: A Colossal Adventure. In *AAAI Conference on Artificial Intelligence*, Vol. 34. 7903–7910.
- Jianliang He, Siyu Chen, Fengzhuo Zhang, and Zhuoran Yang. 2024. From Words to Actions: Unveiling the Theoretical Underpinnings of LLM-Driven Autonomous Systems. (2024). arXiv:2405.19883
- Prannay Hebbar, Yogendra Manawat, Samuel Verboomen, Alesia Ivanova, Selvam Palanimalai, Kunal Bhatia, and Vignesh Baskaran. 2026. SIA: Self Improving AI with Harness & Weight Updates. (2026). arXiv:2605.27276
- Karl Moritz Hermann, Felix Hill, Simon Green, Fumin Wang, Ryan Faulkner, Hubert Soyer, David Szepesvari, Wojciech Marian Czarnecki, Max Jaderberg, and Denis Teplyashin. 2017. Grounded language learning in a simulated 3d world. *arXiv preprint arXiv:1706.06551* (2017).
- Ilgee Hong, Changlong Yu, Liang Qiu, Weixiang Yan, Zhenghao Xu, Haoming Jiang, Qingru Zhang, Qin Lu, Xin Liu, Chao Zhang, and Tuo Zhao. 2025b. Think-RM: Enabling Long-Horizon Reasoning in Generative Reward Models. (2025). arXiv:2505.16265
- Jiseung Hong, Grace Byun, Seungone Kim, Kai Shu, and Jinho D. Choi. 2025a. Measuring Sycophancy of Language Models in Multi-turn Dialogues. arXiv:2505.23840
- Jiwoo Hong, Noah Lee, and James Thorne. 2024a. Orpo: Monolithic preference optimization without reference model. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 11170–11189.
- Sirui Hong, Mingchen Zhuge, Jonathan Chen, Xiwu Zheng, Yuheng Cheng, Jinlin Wang, Ceyao Zhang, Steven Yau, Zijuan Lin, and Liyang Zhou. 2024b. MetaGPT: Meta programming for a multi-agent collaborative framework. In *International Conference on Learning Representations*, Vol. 2024. 23247–23275.

- Nicola Horst, Davide Mazzaccara, Antonia Schmidt, Michael Sullivan, Filippo Momentè, Luca Franceschetti, Philipp Sadler, Sherzod Hakimov, Alberto Testoni, Raffaella Bernardi, Raquel Fernández, Alexander Koller, Oliver Lemon, David Schlangen, Mario Giulianelli, and Alessandro Suglia. 2025. Playpen: An Environment for Exploring Learning Through Conversational Interaction. (2025). arXiv:2504.08590
- Arian Hosseini, Xingdi Yuan, Nikolay Malkin, Aaron Courville, Alessandro Sordani, and Rishabh Agarwal. 2024. V-STaR: Training Verifiers for Self-Taught Reasoners. In *Conference on Language Modeling*. arXiv:2402.06457
- Shengran Hu, Cong Lu, and Jeff Clune. 2024. Automated Design of Agentic Systems. (2024). arXiv:2408.08435
- Tianyu Hu, Zhen Tan, Song Wang, Huaizhi Qu, and Tianlong Chen. 2025b. Multi-Agent Debate for LLM Judges with Adaptive Stability Detection. *arXiv preprint arXiv:2510.12697* (2025). arXiv:2510.12697
- Yuanzhe Hu, Yu Wang, and Julian McAuley. 2025c. Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions. (2025). arXiv:2507.05257
- Zican Hu, Wei Liu, Xiaoye Qu, Xiangyu Yue, Chunlin Chen, Zhi Wang, and Yu Cheng. 2025a. Divide and conquer: Grounding LLMs as efficient decision-making agents via offline hierarchical reinforcement learning. *arXiv preprint arXiv:2505.19761* (2025).
- Hui Huang, Yancheng He, Hongli Zhou, Rui Zhang, Wei Liu, Weixun Wang, Jiaheng Liu, and Wenbo Su. 2025b. Think-J: Learning to Think for Generative LLM-as-a-Judge. (2025). arXiv:2505.14268
- Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Yu, Xinying Song, and Denny Zhou. 2024a. Large language models cannot self-correct reasoning yet. In *International conference on learning representations*, Vol. 2024. 32808–32824.
- Jiaxin Huang, Shixiang Shane Gu, Le Hou, Yuexin Wu, Xuezhi Wang, Hongkun Yu, and Jiawei Han. 2022a. Large Language Models Can Self-Improve. *arXiv preprint arXiv:2210.11610* (2022).
- Jiani Huang, Amish Sethi, Matthew Kuo, Mayank Keoliya, Neelay Velingker, JungHo Jung, Ser-Nam Lim, Ziyang Li, and Mayur Naik. 2025c. ESCA: Contextualizing Embodied Agents via Scene-Graph Generation. arXiv:2510.15963 [cs.CV] <https://arxiv.org/abs/2510.15963>
- Lei Huang, Xiang Cheng, Chenxiao Zhao, Guobin Shen, Junjie Yang, Xiaocheng Feng, Yuxuan Gu, Xing Yu, and Bing Qin. 2026. Bootstrapping Exploration with Group-Level Natural Language Feedback in Reinforcement Learning. (2026). arXiv:2603.04597
- Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. 2023. Voxposer: Composable 3d value maps for robotic manipulation with language models. *arXiv preprint arXiv:2307.05973* (2023).
- Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, and Yevgen Chebotar. 2022b. Inner monologue: Embodied reasoning through planning with language models. *arXiv preprint arXiv:2207.05608* (2022).
- Yue Huang, Yanyuan Chen, Dexuan Xu, Weihua Yue, Huamin Zhang, Meikang Qiu, and Yu Huang. 2025a. MedReflect: Teaching Medical LLMs to Self-Improve via Reflective Correction. *arXiv preprint arXiv:2510.03687* (2025).
- Yuzhen Huang, Weihao Zeng, Kingshan Zeng, Qi Zhu, and Junxian He. 2025d. From Accuracy to Robustness: A Study of Rule- and Model-based Verifiers in Mathematical Reasoning. (2025). arXiv:2505.22203
- Zilin Huang, Zihao Sheng, Yansong Qu, Junwei You, and Sikai Chen. 2024b. VLM-RL: A Unified Vision Language Models and Reinforcement Learning Framework for Safe Autonomous Driving. (2024). arXiv:2412.15544
- Zenan Huang, Yihong Zhuang, Guoshan Lu, Zeyu Qin, Haokai Xu, Tianyu Zhao, Ru Peng, Jiaqi Hu, Zhanming Shen, Xiaomeng Hu, Xijun Gu, Peiyi Tu, Jiaxin Liu, Wenyu Chen, Yuzhuo Fu, Zhiting Fan, Yanmei Gu, Yuanyuan Wang, Zhengkai Yang, Jianguo Li, and Junbo Zhao. 2025e. Reinforcement Learning with Rubric Anchors. (2025). arXiv:2508.12790
- Minyoung Hwang, Joey Hejna, Dorsa Sadigh, and Yonatan Bisk. 2024. MotIF: Motion Instruction Fine-tuning. (2024). arXiv:2409.10683
- Yuki Ito and Qiang Ma. 2025. Ensemble ToT of LLMs and Its Application to Automatic Grading System for Supporting Self-Learning. *arXiv preprint arXiv:2502.16399* (2025).
- Miaomiao Ji, Yanqiu Wu, Zhibin Wu, Shoujin Wang, Jian Yang, Mark Dras, and Usman Naseem. 2025. A Survey on Progress in LLM Alignment from the Perspective of Reward Design. (2025). arXiv:2505.02666
- Shuo Ji, Yibo Li, and Bryan Hooi. 2026. Memory is Reconstructed, Not Retrieved: Graph Memory for LLM Agents. (2026). arXiv:2606.06036
- Dongwei Jiang, Alvin Zhang, Andrew Wang, Nicholas Andrews, and Daniel Khashabi. 2025. Feedback Friction: LLMs Struggle to Fully Incorporate External Feedback. (2025). arXiv:2506.11930
- Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. 2024. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? arXiv:2310.06770 [cs.CL] <https://arxiv.org/abs/2310.06770>
- Frank Joublin, Antonello Ceravola, Pavel Smirnov, Felix Ocker, Joerg Deigmoeller, Anna Belardinelli, Chao Wang, Stephan Hasler, Daniel Tanneberg, and Michael Gienger. 2024. Copal: corrective planning of robot actions with large language models. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 8664–8670.
- Leslie Pack Kaelbling, Michael L Littman, and Anthony R Cassandra. 1998. Planning and acting in partially observable stochastic domains. *Artificial Intelligence* 101, 1–2 (1998), 99–134. doi:10.1016/S0004-3702(98)00023-X
- Ryo Kamoi, Yusen Zhang, Nan Zhang, Jiawei Han, and Rui Zhang. 2024. When can llms actually correct their own mistakes? a critical survey of self-correction of llms. *Transactions of the Association for Computational Linguistics* 12 (2024), 1417–1440.
- Berkcan Kapusuzoglu, Supriyo Chakraborty, Zain Sarwar, Chia-Hsuan Lee, and Sambit Sahu. 2025. Critique-Guided Distillation for Robust Reasoning via Refinement. *arXiv preprint arXiv:2505.11628* (2025). arXiv:2505.11628
- Timo Kaufmann, Paul Weng, Viktor Bengs, and Eyke Hüllermeier. 2023. A survey of reinforcement learning from human feedback. *arXiv preprint arXiv:2312.14925* (2023).
- Manuscript submitted to ACM

- Pei Ke, Bosi Wen, Andrew Feng, Xiao Liu, Xuanyu Lei, Jiale Cheng, Shengyuan Wang, Aohan Zeng, Yuxiao Dong, and Hongning Wang. 2024. Critiquellm: Towards an informative critique generation model for evaluation of large language model generation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 13034–13054.
- Zixuan Ke, Fangkai Jiao, Yifei Ming, Xuan-Phi Nguyen, Austin Xu, Do Xuan Long, Minzhi Li, Chengwei Qin, Peifeng Wang, Silvio Savarese, Caiming Xiong, and Shafiq Joty. 2025. A Survey of Frontiers in LLM Reasoning: Inference Scaling, Learning to Reason, and Agentic Systems. (2025). arXiv:2504.09037
- Zachary Kenton, Noah Y. Siegel, János Kramár, Jonah Brown-Cohen, Samuel Albanie, Jannis Bulian, Rishabh Agarwal, David Lindner, Yunhao Tang, Noah D. Goodman, and Rohin Shah. 2024. On scalable oversight with weak LLMs judging strong LLMs. (2024). arXiv:2407.04622
- Muhammad Khalifa, Rishabh Agarwal, Lajanugen Logeswaran, Jaekyeom Kim, Hao Peng, Moontae Lee, Honglak Lee, and Lu Wang. 2025. Process Reward Models That Think. arXiv:2504.16828 [cs.LG] <https://arxiv.org/abs/2504.16828>
- Akbir Khan, John Hughes, Dan Valentine, Laura Ruis, Kshitij Sachan, Ansh Radhakrishnan, Edward Grefenstette, Samuel R. Bowman, Tim Rocktäschel, and Ethan Perez. 2024. Debating with More Persuasive LLMs Leads to More Truthful Answers. (2024). arXiv:2402.06782
- Omar Khattab, Arnab Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. 2023. DSPy: Compiling Declarative Language Model Calls into Self-Improving Pipelines. (2023). arXiv:2310.03714
- Geunwoo Kim, Pierre Baldi, and Stephen McAleer. 2023. Language Models can Solve Computer Tasks. In *Advances in Neural Information Processing Systems*.
- Jeonghye Kim, Sojeong Rhee, Minbeom Kim, Dohyung Kim, Sangmook Lee, Youngchul Sung, and Kyomin Jung. 2025b. Reflect: World-grounded decision making in llm agents via goal-state reflection. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. 33421–33453.
- Sunghwan Kim, Dongjin Kang, Taeyoon Kwon, Hyungjoo Chae, Jungsoo Won, Dongha Lee, and Jinyoung Yeo. 2024. Evaluating Robustness of Reward Models for Mathematical Reasoning. (2024). arXiv:2410.01729
- Taewoong Kim, Byeonghwi Kim, and Jonghyun Choi. 2025a. Multi-modal grounded planning and efficient replanning for learning embodied agents with a few examples. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 4329–4337.
- Jan Hendrik Kirchner, Yining Chen, Harri Edwards, Jan Leike, Nat McAleese, and Yuri Burda. 2024. Prover-Verifier Games improve legibility of LLM outputs. (2024). arXiv:2407.13692
- Martin Klissarov, Jonathan Cook, Diego Antognini, Hao Sun, Jingling Li, Natasha Jaques, Claudiu Musat, and Edward Grefenstette. 2026. Improving Interactive In-Context Learning from Natural Language Feedback. *arXiv preprint arXiv:2602.16066* (2026).
- Martin Klissarov, Pierluca D’Oro, Shagun Sodhani, Roberta Raileanu, Pierre-Luc Bacon, Pascal Vincent, Amy Zhang, and Mikael Henaff. 2023. MOTIF: Intrinsic Motivation from Artificial Intelligence Feedback. *arXiv preprint arXiv:2310.00166* (2023).
- W Bradley Knox and Peter Stone. 2009. Interactively shaping agents via human reinforcement: The TAMER framework. In *Proceedings of the fifth international conference on Knowledge capture*. 9–16.
- Ryan Koo, Ian Yang, Vipul Raheja, Mingyi Hong, Kwang-Sung Jun, and Dongyeop Kang. 2025. Learning Explainable Dense Reward Shapes via Bayesian Optimization. (2025). arXiv:2504.16272
- Leonard Krasner. 1958. Studies of the conditioning of verbal behavior. *Psychological Bulletin* 55, 3 (1958), 148–170. doi:10.1037/h0040492
- Jakub Grudzien Kuba, Mengting Gu, Qi Ma, Yuandong Tian, Vijai Mohan, and Jason Chen. 2025. Language Self-Play For Data-Free Training. (2025). arXiv:2509.07414
- Aviral Kumar, Vincent Zhuang, Rishabh Agarwal, Yi Su, John D Co-Reyes, Avi Singh, Kate Baumli, Shariq Iqbal, Colton Bishop, and Rebecca Roelofs. 2024. Training language models to self-correct via reinforcement learning. *arXiv preprint arXiv:2409.12917* (2024).
- Peng Lai, Zhihao Ou, Yong Wang, Longyue Wang, Jian Yang, Yun Chen, and Guanhua Chen. 2026. BiasScope: Towards Automated Detection of Bias in LLM-as-a-Judge Evaluation. (2026). arXiv:2602.09383
- Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James V Miranda, Alisa Liu, Nouha Dziri, and Shane Lyu. 2024a. Tulu 3: Pushing frontiers in open language model post-training. *arXiv preprint arXiv:2411.15124* (2024).
- Nathan Lambert, Valentina Pyatkin, Jacob Morrison, LJ Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, Noah A. Smith, and Hannaneh Hajishirzi. 2024b. RewardBench: Evaluating Reward Models for Language Modeling. (2024). arXiv:2403.13787
- Tian Lan, Wenwei Zhang, Chengqi Lyu, Shuaibin Li, Chen Xu, Heyan Huang, Dahua Lin, Xian-Ling Mao, and Kai Chen. 2024a. Training Language Models to Critique With Multi-agent Feedback. (2024). arXiv:2410.15287
- Tian Lan, Wenwei Zhang, Chen Xu, Heyan Huang, Dahua Lin, Kai Chen, and Xian-ling Mao. 2024b. CriticEval: Evaluating Large Language Model as Critic. (2024). arXiv:2402.13764
- Hao Lang, Fei Huang, and Yongbin Li. 2025. Debate Helps Weak-to-Strong Generalization. (2025). arXiv:2501.13124
- Harrison Lee, Samrat Phatale, Hassan Mansoor, Thomas Mesnard, Johan Ferret, Kellie Lu, Colton Bishop, Ethan Hall, Victor Carbune, and Abhinav Rastogi. 2023. Rlaif vs. rlhf: Scaling reinforcement learning from human feedback with ai feedback. *arXiv preprint arXiv:2309.00267* (2023).
- Lisa Lee, Reilly Boen, and Chelsea Finn. 2025a. Feedback Descent: Open-Ended Text Optimization via Pairwise Comparison. *arXiv preprint* (2025).
- Tony Lee, Andrew Wagenmaker, Karl Pertsch, Percy Liang, Sergey Levine, and Chelsea Finn. 2026. RoboReward: General-Purpose Vision-Language Reward Models for Robotics. (2026). arXiv:2601.00675
- Yu-Ang Lee, Guan-Ting Yi, Mei-Yi Liu, Jui-Chao Lu, Guan-Bo Yang, and Yun-Nung Chen. 2025b. Compound AI Systems Optimization: A Survey of Methods, Challenges, and Future Directions. (2025). arXiv:2506.08234

- Andrew Li, Toryn Klassen, Andrew Wang, Parand A Alamdari, and Sheila McIlraith. 2026a. Ground-Compose-Reinforce: Grounding Language in Agentic Behaviours using Limited Data. *Advances in Neural Information Processing Systems* 38 (2026), 160838–160874.
- Dawei Li, Bohan Jiang, Liangjie Huang, Alimohammad Beigi, Chengshuai Zhao, Zhen Tan, Amrita Bhattacharjee, Yuxuan Jiang, Canyu Chen, Tianhao Wu, Kai Shu, Lu Cheng, and Huan Liu. 2024c. From Generation to Judgment: Opportunities and Challenges of LLM-as-a-judge. (2024). arXiv:2411.16594
- Haitao Li, Qian Dong, Junjie Chen, Huixue Su, Yujia Zhou, Qingyao Ai, Ziyi Ye, and Yiqun Liu. 2024a. LLMs-as-Judges: A Comprehensive Survey on LLM-based Evaluation Methods. (2024). arXiv:2412.05579
- Jiahui Li, Lin Li, Tai-wei Chang, Kun Kuang, Long Chen, Jun Zhou, and Cheng Yang. 2024d. RED: Unleashing Token-Level Rewards from Holistic Feedback via Reward Redistribution. (2024). arXiv:2411.08302
- Jiaze Li, Hao Yin, Wenhui Tan, Jingyang Chen, Boshen Xu, Yuxun Qu, Yijing Chen, Jianzhong Ju, Zhenbo Luo, and Jian Luan. 2025b. REVISOR: Beyond Textual Reflection, Towards Multimodal Introspective Reasoning in Long-Form Video Understanding. *arXiv preprint* (2025).
- Songze Li, Chuokun Xu, Jiaying Wang, Xueluan Gong, Chen Chen, Jirui Zhang, Jun Wang, Kwok-Yan Lam, and Shouling Ji. 2025a. LLMs Cannot Reliably Judge (Yet?): A Comprehensive Assessment on the Robustness of LLM-as-a-Judge. arXiv:2506.09443
- Sunzhu Li, Jiale Zhao, Miteto Wei, Huimin Ren, Yang Zhou, Jingwen Yang, Shunyu Liu, Kaike Zhang, and Wei Chen. 2026b. RubricHub: A Comprehensive and Highly Discriminative Rubric Dataset via Automated Coarse-to-Fine Generation. (2026). arXiv:2601.08430
- Xinzhe Li. 2025. A Survey on LLM Test-Time Compute via Search: Tasks, LLM Profiling, Search Algorithms, and Relevant Frameworks. (2025). arXiv:2501.10069
- Yunxuan Li, Yibing Du, Jiageng Zhang, Le Hou, Peter Grabowski, Yeqing Li, and Eugene Ie. 2024b. Improving multi-agent debate with sparse communication topology. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. 7281–7294.
- Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. 2022. Code as policies: Language model programs for embodied control, 2023. URL <https://arxiv.org/abs/2209.07753> 3 (2022).
- Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Shuming Shi, and Zhaopeng Tu. 2024a. Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate. arXiv:2305.19118 [cs.CL] <https://arxiv.org/abs/2305.19118>
- Xun Liang, Shichao Song, Zifan Zheng, Hanyu Wang, Qingchen Yu, Xunkai Li, Rong-Hua Li, Yi Wang, Zhonghao Wang, Feiyu Xiong, and Zhiyu Li. 2024b. Internal Consistency and Self-Feedback in Large Language Models: A Survey. (2024). arXiv:2407.14507
- Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let’s verify step by step. In *International Conference on Learning Representations*, Vol. 2024. 39578–39601. arXiv:2305.20050
- Huawei Lin, Peng Li, Jie Song, Fuxin Jiang, and Tiejing Zhang. 2026. MUSE-Autoskill: Self-Evolving Agents via Skill Creation, Memory, Management, and Evaluation. (2026). arXiv:2605.27366
- Mathieu Lin, Jenny Sheng, Andrew Zhao, Shenzhi Wang, Yang Yue, Victor Shea Jay Huang, Huan Liu, Jun Liu, Gao Huang, and Yong-Jin Liu. 2024b. Scaffolded Language Models with Language Supervision for Mixed-Autonomy: A Survey. (2024). arXiv:2410.16392
- Muhan Lin, Shuyang Shi, Yue Guo, Vaishnav Tadiparthi, Behdad Chalaki, Ehsan Moradi Pari, Simon Stepputtis, Woojun Kim, Joseph Campbell, and Katia Sycara. 2025b. Speaking the Language of Teamwork: LLM-Guided Credit Assignment in Multi-Agent Reinforcement Learning. (2025). arXiv:2502.03723
- Zijun Lin, Jiafei Duan, Haoquan Fang, Dieter Fox, Ranjay Krishna, Cheston Tan, and Bihan Wen. 2025a. FailSafe: Reasoning and Recovery from Failures in Vision-Language-Action Models. (2025). arXiv:2510.01642
- Zicheng Lin, Zhibin Gou, Tian Liang, Ruilin Luo, Haowei Liu, and Yujiu Yang. 2024a. Criticbench: Benchmarking llms for critique-correct reasoning. In *Findings of the Association for Computational Linguistics: ACL 2024*. 1552–1587.
- Zijie Lin and Bryan Hooi. 2025. Enhancing Multi-Agent Debate System Performance via Confidence Expression. (2025). arXiv:2509.14034
- Chenxi Liu, Yanshuo Chen, Ruibo Chen, Tianyi Xiong, Tong Zheng, and Heng Huang. 2026a. Learning from Self-Debate: Preparing Reasoning Models for Multi-Agent Debate. arXiv:2601.22297 [cs.CL] <https://arxiv.org/abs/2601.22297>
- Hao Liu, Carmelo Sferrazza, and Pieter Abbeel. 2023. Chain of Hindsight Aligns Language Models with Feedback. *arXiv preprint arXiv:2302.02676* (2023).
- Hongyi Liu, Haoyan Yang, Tao Jiang, Bo Tang, Feiyu Xiong, Yuyu Luo, and Zhiyu Li. 2026f. SkillsVote: Lifecycle Governance of Agent Skills from Collection, Recommendation to Evolution. (2026). arXiv:2605.18401
- Pangpang Liu, Chengchun Shi, and Will Wei Sun. 2026c. Reinforcement Learning from Human Feedback: A Statistical Perspective. (2026). arXiv:2604.02507
- Qingshan Liu, Guoqing Wang, Wen Wu, Jingqi Huang, Xinqi Tao, Dejie Song, Jie Zhou, and Liang He. 2026d. MemPro: Agentic Memory Systems as Evolvable Programs. (2026). arXiv:2606.00619
- Qiyuan Liu, Hao Xu, Xuhong Chen, Wei Chen, Yee Whye Teh, and Ning Miao. 2025c. Enhancing Large Language Model Reasoning with Reward Models: An Analytical Survey. (2025). arXiv:2510.01925
- Qihao Liu, Luoxin Ye, Wufei Ma, Yu-Cheng Chou, and Alan Yuille. 2025e. Generative Adversarial Reasoner: Enhancing LLM Reasoning with Adversarial Reinforcement Learning. *arXiv preprint arXiv:2512.16917* (2025).
- ShaoZhen Liu, Xinting Huang, Houwen Peng, Xin Chen, Xinyang Song, Qi Li, and Zhenan Sun. 2026b. Dual-Phase LLM Reasoning: Self-Evolved Mathematical Frameworks. *arXiv preprint arXiv:2601.05616* (2026).
- Shiyan Liu, Qifeng Xia, Qiyun Xia, Yisheng Liu, Xinyu Yu, and Rui Qu. 2026e. Reflection in the Dark: Exposing and Escaping the Black Box in Reflective Prompt Optimization. (2026). arXiv:2603.18388
- Tianci Liu, Ran Xu, Tony Yu, Ilgee Hong, Carl Yang, Tuo Zhao, and Haoyu Wang. 2025d. OpenRubrics: Towards Scalable Synthetic Rubric Generation for Reward Modeling and LLM Alignment. (2025). arXiv:2510.07743

- Xiaoqian Liu, Ke Wang, Yuchuan Wu, Fei Huang, Yongbin Li, Junge Zhang, and Jianbin Jiao. 2025a. Agentic Reinforcement Learning with Implicit Step Rewards. *arXiv:2509.19199* [cs.CL] <https://arxiv.org/abs/2509.19199>
- Yantao Liu, Zijun Yao, Rui Min, Yixin Cao, Lei Hou, and Juanzi Li. 2024. RM-Bench: Benchmarking Reward Models of Language Models with Subtlety and Style. (2024). *arXiv:2410.16184*
- Zijun Liu, Peiyi Wang, Runxin Xu, Shirong Ma, Chong Ruan, Peng Li, Yang Liu, and Yu Wu. 2025b. Inference-Time Scaling for Generalist Reward Modeling. (2025). *arXiv:2504.02495*
- Saüc Abadal Lloret, Shehzaad Dhuliawala, Keerthiram Murugesan, and Mrinmaya Sachan. 2024. Towards aligning language models with textual feedback. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 20240–20266.
- Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. 2024. The ai scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292* (2024).
- Guanxing Lu, Wenkai Guo, Chubin Zhang, Yuheng Zhou, Haonan Jiang, Zifeng Gao, Yansong Tang, and Ziwei Wang. 2025. VLA-RL: Towards Masterful and General Robotic Manipulation with Scalable Reinforcement Learning. (2025). *arXiv:2505.18719*
- Xinyu Lu, Kaiqi Zhang, Jinglin Yang, Boxi Cao, Yaojie Lu, Hongyu Lin, Min He, Xianpei Han, and Le Sun. 2026. P²O: Joint Policy and Prompt Optimization. (2026). *arXiv:2603.21877*
- Jelena Luketina, Nantas Nardelli, Gregory Farquhar, Jakob Foerster, Jacob Andreas, Edward Grefenstette, Shimon Whiteson, and Tim Rocktäschel. 2019. A survey of reinforcement learning informed by natural language. *arXiv preprint arXiv:1906.03926* (2019).
- Liangchen Luo, Yinxiao Liu, Rosanne Liu, Samrat Phatale, Meiqi Guo, Harsh Lara, Yunxuan Li, Lei Shu, Yun Zhu, and Lei Meng. 2024. Improve mathematical reasoning in language models by automated process supervision. *arXiv preprint arXiv:2406.06592* (2024).
- Renjie Luo, Zichen Liu, Xiangyan Liu, Chao Du, Min Lin, Wenhui Chen, Wei Lu, and Tianyu Pang. 2025. Language models can learn from verbal feedback without scalar rewards. *arXiv preprint arXiv:2509.22638* (2025).
- Corey Lynch and Pierre Sermanet. 2021. Language Conditioned Imitation Learning over Unstructured Data. In *Robotics: Science and Systems*.
- Xing Han Lü, Amirhossein Kazemnejad, Nicholas Meade, Arkil Patel, Dongchan Shin, Alejandra Zambrano, Karolina Stańczak, Peter Shaw, Christopher J. Pal, and Siva Reddy. 2025. AgentRewardBench: Evaluating Automatic Evaluations of Web Agent Trajectories. (2025). *arXiv:2504.08942*
- Qiyao Ma, Dechen Gao, Rui Cai, Boqi Zhao, Hanchu Zhou, Junshan Zhang, and Zhe Zhao. 2026. Personalized RewardBench: Evaluating Reward Models with Human Aligned Personalization. (2026). *arXiv:2604.07343*
- Ruotian Ma, Xiaolei Wang, Xin Zhou, Jian Li, Nan Du, Tao Gui, Qi Zhang, and Xuanjing Huang. 2024c. Are Large Language Models Good Prompt Optimizers? (2024). *arXiv:2402.02101*
- Yecheng Jason Ma, William Liang, Guanzhi Wang, De-An Huang, Osbert Bastani, Dinesh Jayaraman, Yuke Zhu, and Jim Fan. 2024a. Eureka: Human-level reward design via coding large language models. In *International conference on learning Representations*, Vol. 2024. 26516–26560.
- Yecheng Jason Ma, William Liang, Hung-Ju Wang, Sam Wang, Yuke Zhu, Linxi Fan, Osbert Bastani, and Dinesh Jayaraman. 2024b. DrEureka: Language Model Guided Sim-To-Real Transfer. (2024). *arXiv:2406.01967*
- James MacGlashan, Mark K Ho, Robert Loftin, Bei Peng, Guan Wang, David L Roberts, Matthew E Taylor, and Michael L Littman. 2017. Interactive learning from policy-dependent human feedback. In *International conference on machine learning*. PMLR, 2285–2294. *arXiv:1701.06049*
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems*.
- Anas Mahmoud, MohammadHossein Rezaei, Zihao Wang, Anisha Gunjal, Bing Liu, and Yunzhong He. 2026. Reward Hacking in Rubric-Based Reinforcement Learning. (2026). *arXiv:2605.12474*
- Saumya Malik, Valentina Pyatkin, Sander Land, Jacob Morrison, Noah A. Smith, Hannaneh Hajishirzi, and Nathan Lambert. 2025. RewardBench 2: Advancing Reward Model Evaluation. (2025). *arXiv:2506.01937*
- Lars Malmqvist. 2024. Sycophancy in Large Language Models: Causes and Mitigations. *arXiv:2411.15287*
- Narek Maloyan and Dmitry Namiot. 2025. Adversarial Attacks on LLM-as-a-Judge Systems: Insights from Prompt Injections. *arXiv:2504.18333*
- Nat McAleese, Rai Michael Pokorny, Juan Felipe Ceron Uribe, Evgenia Nitishinskaya, Maja Trebacz, and Jan Leike. 2024. LLM Critics Help Catch LLM Bugs. *arXiv preprint arXiv:2407.00215* (2024).
- Sabrina McCallum, Max Taylor-Davies, Stefano Albrecht, and Alessandro Suglia. 2023. Is Feedback All You Need? Leveraging Natural Language Feedback in Goal-Conditioned Reinforcement Learning. In *NeurIPS 2023 Workshop on Goal-Conditioned Reinforcement Learning*. *arXiv:2312.04736*
- Lingrui Mei, Jiayu Yao, Yuyao Ge, Yiwei Wang, Baolong Bi, Yujun Cai, Jiazhi Liu, Mingyu Li, Zhong-Zhi Li, Duzhen Zhang, Chenlin Zhou, Jiayi Mao, Tianze Xia, Jiafeng Guo, and Shenghua Liu. 2025. A Survey of Context Engineering for Large Language Models. (2025). *arXiv:2507.13334*
- Alexander Meinke, Bronson Schoen, Jérémy Scheurer, Mikita Balesni, Rusheb Shah, and Marius Hobbhahn. 2024. Frontier models are capable of in-context scheming. *arXiv preprint arXiv:2412.04984* (2024).
- Yu Meng, Mengzhou Xia, and Danqi Chen. 2024. Simpo: Simple preference optimization with a reference-free reward. *Advances in Neural Information Processing Systems* 37 (2024), 124198–124235.
- Azalia Mirhoseini, Anna Goldie, Mustafa Yazgan, Joe Wenjie Jiang, Ebrahim Songhori, Shen Wang, Young-Joon Lee, Eric Johnson, Omkar Pathak, and Azade Nova. 2021. A graph placement methodology for fast chip design. *Nature* 594, 7862 (2021), 207–212.
- Dipendra Misra, John Langford, and Yoav Artzi. 2017. Mapping instructions and visual observations to actions with reinforcement learning. In *Proceedings of the 2017 conference on empirical methods in natural language processing*. 1004–1015.

- Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A Rusu, Joel Veness, Marc G Bellemare, Alex Graves, Martin Riedmiller, Andreas K Fiedjeland, and Georg Ostrovski. 2015. Human-level control through deep reinforcement learning. *Nature* 518, 7540 (2015), 529–533.
- Ruaridh Mon-Williams, Gen Li, Ran Long, Wenqian Du, and Christopher G Lucas. 2025. Embodied large language models enable robots to complete complex tasks in unpredictable environments. *Nature Machine Intelligence* 7, 4 (2025), 592–601.
- Sumeet Ramesh Motwani, Chandler Smith, Rocktim Jyoti Das, Rafael Rafailov, Ivan Laptev, Philip H. S. Torr, Fabio Pizzati, Ronald Clark, and Christian Schroeder de Witt. 2024. MALT: Improving Reasoning with Multi-Agent LLM Training. (2024). [arXiv:2412.01928](#)
- Vaskar Nath, Pranav Raja, Claire Yoon, and Sean Hendryx. 2025. ToolComp: A Multi-Tool Reasoning & Process Supervision Benchmark. (2025). [arXiv:2501.01290](#)
- Manh Nguyen, Anh Nguyen, Dung Nguyen, Svetha Venkatesh, and Hung Le. 2026. Hear Both Sides: Efficient Multi-Agent Debate via Diversity-Aware Message Retention. (2026). [arXiv:2603.20640](#)
- Allen Nie, Ching-An Cheng, Andrey Kolobov, and Adith Swaminathan. 2024. The Importance of Directional Feedback for LLM-based Optimizers. (2024). [arXiv:2405.16434](#)
- Xuying Ning, Katherine Tieu, Dongqi Fu, Tianxin Wei, Zihao Li, Yuanchen Bei, Jiaru Zou, Mengting Ai, Zhining Liu, and Ting-Wei Li. 2026. Code as Agent Harness. *arXiv preprint arXiv:2605.18747* (2026).
- Theo X Olausson, Jeevana Priya Inala, Chenglong Wang, Jianfeng Gao, and Armando Solar-Lezama. 2024. Is self-repair a silver bullet for code generation?. In *International Conference on Learning Representations*, Vol. 2024. 36545–36593.
- Krista Opsahl-Ong, Michael J Ryan, Josh Purtell, David Broman, Christopher Potts, Matei Zaharia, and Omar Khattab. 2024. Optimizing Instructions and Demonstrations for Multi-Stage Language Model Programs. In *Conference on Empirical Methods in Natural Language Processing*.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, and Alex Ray. 2022. Training language models to follow instructions with human feedback. *Advances in neural information processing systems* 35 (2022), 27730–27744.
- Siru Ouyang, Jun Yan, I-Hung Hsu, Yanfei Chen, Ke Jiang, Zifeng Wang, Rujun Han, Long T. Le, Samira Daruki, Xiangru Tang, Vishy Tirumalashetty, George Lee, Mahsan Rofouei, Hangfei Lin, Jiawei Han, Chen-Yu Lee, and Tomas Pfister. 2025. ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory. (2025). [arXiv:2509.25140](#)
- Paul Pacaud, Ricardo Garcia, Shizhe Chen, and Cordelia Schmid. 2025. Scaling Cross-Environment Failure Reasoning Data for Vision-Language Robotic Manipulation. (2025). [arXiv:2512.01946](#)
- Alexander Pan, Erik Jones, Meena Jagadeesan, and Jacob Steinhardt. 2024. Feedback Loops With Language Models Drive In-Context Reward Hacking. [arXiv:2402.06627](#)
- Liangming Pan, Michael Saxon, Wenda Xu, Deepak Nathani, Xinyi Wang, and William Yang Wang. 2023. Automatically Correcting Large Language Models: Surveying the landscape of diverse self-correction strategies. (2023). [arXiv:2308.03188](#)
- Manav Pandey. 2026. LLMs Know They’re Wrong and Agree Anyway: The Shared Sycophancy-Lying Circuit. [arXiv:2604.19117](#)
- Henry Papadatos and Rachel Freedman. 2024. Linear Probe Penalties Reduce LLM Sycophancy. [arXiv:2412.00967](#)
- Chanwoo Park, Seungju Han, Xingzhi Guo, Asuman Ozdaglar, Kaiqing Zhang, and Joo-Kyung Kim. 2025. MAPoRL: Multi-Agent Post-Co-Training for Collaborative Large Language Models with Reinforcement Learning. (2025). [arXiv:2502.18439](#)
- Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *ACM Symposium on User Interface Software and Technology (UIST)*.
- Pankayaraj Pathmanathan, Souradip Chakraborty, Xiangyu Liu, Yongyuan Liang, and Furong Huang. 2024. Is poisoning a real threat to LLM alignment? Maybe more so than you think. [arXiv:2406.12091](#)
- Debjit Paul, Mete Ismayilzada, Maxime Peyrard, Beatriz Borges, Antoine Bosselut, Robert West, and Boi Faltings. 2024. REFINER: Reasoning Feedback on Intermediate Representations. *arXiv preprint arXiv:2304.01904* (2024).
- Baolin Peng, Michel Galley, Pengcheng He, Hao Cheng, Yujia Xie, Yu Hu, Qiuyuan Huang, Lars Liden, Zhou Yu, Weizhu Chen, and Jianfeng Gao. 2023. Check Your Facts and Try Again: Improving Large Language Models with External Knowledge and Automated Feedback. *arXiv preprint arXiv:2302.12813* (2023).
- Eduardo Pignatelli, Johan Ferret, Tim Rockäschel, Edward Grefenstette, Davide Paglieri, Samuel Coward, and Laura Toni. 2024. Assessing the Zero-Shot Capabilities of LLMs for Action Evaluation in RL. (2024). [arXiv:2409.12798](#)
- Reid Pryzant, Dan Iter, Jerry Li, Yin Tat Lee, Chenguang Zhu, and Michael Zeng. 2023. Automatic Prompt Optimization with “Gradient Descent” and Beam Search. In *Conference on Empirical Methods in Natural Language Processing*.
- Moschoula Pternea, Prerna Singh, Abir Chakraborty, Yagna Oruganti, Mirco Milletari, Sayli Bapat, and Kebei Jiang. 2024. The rl/llm taxonomy tree: Reviewing synergies between reinforcement learning and large language models. *Journal of Artificial Intelligence Research* 80 (2024), 1525–1573.
- Keyang Qian, Yixin Cheng, Rui Guan, Wei Dai, Flora Jin, Kaixun Yang, Sadia Nawaz, Zachari Swiecki, Guanliang Chen, and Lixiang Yan. 2025. Dean of LLM Tutors: A Framework for Automated Quality Review of AI-generated Feedback. *arXiv preprint arXiv:2508.05952* (2025).
- Yun Qu, Yuhang Jiang, Boyuan Wang, Yixiu Mao, Cheems Wang, Chang Liu, and Xiangyang Ji. 2024a. Latent Reward: LLM-Empowered Credit Assignment in Episodic Reinforcement Learning. (2024). [arXiv:2412.11120](#)
- Yuxiao Qu, Tianjun Zhang, Naman Garg, and Aviral Kumar. 2024b. Recursive Introspection: Teaching Language Model Agents How to Self-Improve. *arXiv preprint arXiv:2407.18219* (2024).

Manuscript submitted to ACM

- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems* 36 (2023), 53728–53741.
- Salman Rahman, Sruthi Gorantla, Arpit Gupta, Swastik Roy, Nanyun Peng, and Yang Liu. 2025. SPARK: Stepwise Process-Aware Rewards for Reference-Free Reinforcement Learning. (2025). [arXiv:2512.03244](#)
- Vyas Raina, Adian Liusie, and Mark Gales. 2024. Is LLM-as-a-Judge Robust? Investigating Universal Adversarial Attacks on Zero-shot LLM Assessment. [arXiv:2402.14016](#)
- Kiran Ramnath, Kang Zhou, Sheng Guan, Soumya Smruti Mishra, Xuan Qi, Zhengyuan Shen, Shuai Wang, Sangmin Woo, Sullam Jeoung, Yawei Wang, Haozhu Wang, Han Ding, Yuzhe Lu, Zhichao Xu, Yun Zhou, Balasubramaniam Srinivasan, Qiaojing Yan, Yueyan Chen, Haibo Ding, Panpan Xu, and Lin Lee Cheong. 2025. A Systematic Survey of Automatic Prompt Optimization Techniques. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 33066–33098. [doi:10.18653/v1/2025.emnlp-main.1681](#)
- MohammadHossein Rezaei, Robert Vacareanu, Zihao Wang, Clinton Wang, Bing Liu, Yunzhong He, and Afra Feyza Akyürek. 2025. Online Rubrics Elicitation from Pairwise Comparisons. (2025). [arXiv:2510.07284](#)
- Amit Roth, Ankur Samanta, Matan Halevy, Yoav Levine, and Yonathan Efroni. 2026. Hack-Verifiable Environments: Towards Evaluating Reward Hacking at Scale. [arXiv:2605.20744](#)
- Swarnadeep Saha, Xian Li, Marjan Ghazvininejad, Jason Weston, and Tianlu Wang. 2025. Learning to Plan & Reason for Evaluation with Thinking-LLM-as-a-Judge. (2025). [arXiv:2501.18099](#)
- Yusuke Sakai, Hidetaka Kamigaito, and Taro Watanabe. 2025. Revisiting compositional generalization capability of large language models considering instruction following ability. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 31219–31238.
- Bidipta Sarkar, Warren Xia, C. Karen Liu, and Dorsa Sadigh. 2025. Training Language Models for Social Deduction with Multi-Agent Reinforcement Learning. (2025). [arXiv:2502.06060](#)
- William Saunders, Catherine Yeh, Jeff Wu, Steven Bills, Long Ouyang, Jonathan Ward, and Jan Leike. 2022. Self-critiquing models for assisting human evaluators. *arXiv preprint arXiv:2206.05802* (2022).
- Jérémy Scheurer, Jon Ander Campos, Tomasz Korbak, Jun Shern Chan, Angelica Chen, Kyunghyun Cho, and Ethan Perez. 2023. Training language models with language feedback at scale. *arXiv preprint arXiv:2303.16755* (2023).
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. *Advances in neural information processing systems* 36 (2023), 68539–68551.
- Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Michael Moor, Zicheng Liu, and Emad Barsoum. 2025. Agent laboratory: Using llm agents as research assistants. *Findings of the Association for Computational Linguistics: EMNLP 2025* (2025), 5977–6043.
- Sheila Schoepp, Masoud Jafaripour, Yingyue Cao, Tianpei Yang, Fatemeh Abdollahi, Shadan Golestan, Zahin Sufiyan, Osmar R. Zaiane, and Matthew E. Taylor. 2025. The Evolving Landscape of LLM- and VLM-Integrated Reinforcement Learning. (2025). [arXiv:2502.15214](#)
- Philip Schroeder, Thomas Weng, Karl Schmeckpeper, Eric Rosen, Stephen Hart, and Ondrej Biza. 2026. SOLE-R1: Video-Language Reasoning as the Sole Reward for On-Robot Reinforcement Learning. (2026). [arXiv:2603.28730](#)
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347* (2017).
- Leo Schwinn, Moritz Ladenburger, Tim Beyer, Mehrnaz Mofakhami, Gauthier Gidel, and Stephan Günnemann. 2026. A Coin Flip for Safety: LLM Judges Fail to Reliably Measure Adversarial Robustness. [arXiv:2603.06594](#)
- Andrew W Senior, Richard Evans, John Jumper, James Kirkpatrick, Laurent Sifre, Tim Green, Chongli Qin, Augustin Zidek, Alexander WR Nelson, and Alex Bridgland. 2020. Improved protein structure prediction using potentials from deep learning. *Nature* 577, 7792 (2020), 706–710.
- Mrinank Sharma, Meg Tong, Tomek Korbak, David Duvenaud, Amanda Askeel, Sam Bowman, Esin Durmus, Zac Hatfield-Dodds, Scott Johnston, and Shauna Kravec. 2024. Towards understanding sycophancy in language models. In *International Conference on Learning Representations*, Vol. 2024. 110–144.
- Pratyusha Sharma, Balakumar Sundaralingam, Valts Blukis, Chris Paxton, Tucker Hermans, Antonio Torralba, Jacob Andreas, and Dieter Fox. 2022. Correcting robot plans with natural language feedback. *arXiv preprint arXiv:2204.05186* (2022).
- Haizhou Shi, Ye Liu, Bo Pang, Zeyu Leo Liu, Hao Wang, Silvio Savarese, Caiming Xiong, Yingbo Zhou, and Semih Yavuz. 2025a. SSR: Socratic Self-Refine for Large Language Model Reasoning. *arXiv preprint arXiv:2511.10621* (2025).
- Jiawen Shi, Zenghui Yuan, Yinuo Liu, Yue Huang, Pan Zhou, Lichao Sun, and Neil Zhenqiang Gong. 2024b. Optimization-based Prompt Injection Attack to LLM-as-a-Judge. [arXiv:2403.17710](#)
- Jiawen Shi, Zenghui Yuan, Guiyao Tie, Pan Zhou, Neil Zhenqiang Gong, and Lichao Sun. 2025b. Prompt injection attack to tool selection in llm agents. *arXiv preprint arXiv:2504.19793* (2025).
- Lucy Xiaoyang Shi, Zheyuan Hu, Tony Z. Zhao, Archit Sharma, Karl Pertsch, Jianlan Luo, Sergey Levine, and Chelsea Finn. 2024a. Yell At Your Robot: Improving On-the-Fly from Language Corrections. (2024). [arXiv:2403.12910](#)
- Noah Shinn, Federico Cassano, Berman Labash, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. In *Advances in Neural Information Processing Systems*, Vol. 36.
- Kumar Shridhar, Alessandro Stolfo, and Mrinmaya Sachan. 2023. Distilling Reasoning Capabilities into Smaller Language Models. In *Findings of the Association for Computational Linguistics: ACL*.

- Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. 2020. Alfvorld: Aligning text and embodied environments for interactive learning. *arXiv preprint arXiv:2010.03768* (2020).
- Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. 2023. The curse of recursion: Training on generated data makes models forget. *arXiv preprint arXiv:2305.17493* (2023).
- David Silver, Aja Huang, Chris J Maddison, Arthur Guez, Laurent Sifre, George Van Den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, and Marc Lanctot. 2016. Mastering the game of Go with deep neural networks and tree search. *nature* 529, 7587 (2016), 484–489.
- Avi Singh, John D Co-Reyes, Rishabh Agarwal, Ankesh Anand, Piyush Patil, Xavier Garcia, Peter J Liu, James Harrison, Jaehoon Lee, and Kelvin Xu. 2023. Beyond human data: Scaling self-training for problem-solving with language models. *arXiv preprint arXiv:2312.06585* (2023).
- Ishika Singh, Valts Blukis, Arsalan Mousavian, Ankit Goyal, Danfei Xu, Jonathan Tremblay, Dieter Fox, Jesse Thomason, and Animesh Garg. 2022. Progprompt: Generating situated robot task plans using large language models. *arXiv preprint arXiv:2209.11302* (2022).
- Prasann Singhal, Tanya Goyal, Jiacheng Xu, and Greg Durrett. 2023. A long way to go: Investigating length correlations in RLHF. *arXiv preprint arXiv:2310.03716* (2023).
- Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. 2024. Scaling LLM Test-Time Compute Optimally Can Be More Effective than Scaling Model Parameters. *arXiv preprint arXiv:2408.03314* (2024).
- Guijin Son, Dongkeun Yoon, Juyoung Suk, Javier Aula-Blasco, Mano Aslan, Vu Trong Kim, Shayekh Bin Islam, Jaume Prats-Cristià, Lucía Tormo-Bañuelos, and Seungone Kim. 2024. MM-Eval: A Multilingual Meta-Evaluation Benchmark for LLM-as-a-Judge and Reward Models. (2024). [arXiv:2410.17578](#)
- Mingyang Song, Zhaochen Su, Xiaoye Qu, Jiawei Zhou, and Yu Cheng. 2025. PRMBench: A Fine-grained and Challenging Benchmark for Process-Level Reward Models. (2025). [arXiv:2501.03124](#)
- Yuda Song, Lili Chen, Fahim Tajwar, Remi Munos, Deepak Pathak, J Andrew Bagnell, Aarti Singh, and Andrea Zanette. 2026. Expanding the Capabilities of Reinforcement Learning via Text Feedback. *arXiv preprint arXiv:2602.02482* (2026).
- Saksham Sahai Srivastava and Vaneet Aggarwal. 2025. A Technical Survey of Reinforcement Learning Techniques for Large Language Models. (2025). [arXiv:2507.04136](#)
- Elias Stengel-Eskin, Peter Hase, and Mohit Bansal. 2024. Teaching Models to Balance Resisting and Accepting Persuasion. (2024). [arXiv:2410.14596](#)
- Moritz Stephan, Alexander Khazatsky, Eric Mitchell, Annie S Chen, Sheryl Hsu, Archit Sharma, and Chelsea Finn. 2024. RLVF: Learning from Verbal Feedback without Overgeneralization. (2024). [arXiv:2402.10893](#)
- Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F Christiano. 2020. Learning to summarize with human feedback. *Advances in neural information processing systems* 33 (2020), 3008–3021. [arXiv:2009.01325](#)
- Alexander L Strehl, Lihong Li, and Michael L Littman. 2009. Reinforcement Learning in Finite MDPs: PAC Analysis. *Journal of Machine Learning Research* 10, 84 (2009), 2413–2444.
- Haoran Su, Yandong Sun, and Congjia Yu. 2026. The End of Reward Engineering: How LLMs Are Redefining Multi-Agent Coordination. (2026). [arXiv:2601.08237](#)
- Vighnesh Subramaniam, Yilun Du, Joshua B. Tenenbaum, Antonio Torralba, Shuang Li, and Igor Mordatch. 2025. Multiagent Finetuning: Self Improvement with Diverse Reasoning Chains. (2025). [arXiv:2501.05707](#)
- Aleksa Sukovic and Goran Radanovic. 2024. Reward Design for Justifiable Sequential Decision-Making. (2024). [arXiv:2402.15826](#)
- Theodore R. Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. 2023. Cognitive Architectures for Language Agents. (2023). [arXiv:2309.02427](#)
- Haotian Sun, Yuchen Zhuang, Lingkai Kong, Bo Dai, and Chao Zhang. 2023. AdaPlanner: Adaptive Planning from Feedback with Language Models. In *Advances in Neural Information Processing Systems*.
- Shengjie Sun, Runze Liu, Jiafei Lyu, Jing-Wen Yang, Liangpeng Zhang, and Xiu Li. 2025a. A large language model-driven reward design framework via dynamic feedback for reinforcement learning. *Knowledge-Based Systems* 326 (2025), 114065.
- Shengjie Sun, Jiafei Lyu, Runze Liu, Mengbei Yan, Bo Liu, Deheng Ye, and Xiu Li. 2025b. PROF: An LLM-based Reward Code Preference Optimization Framework for Offline Imitation Learning. *arXiv preprint arXiv:2511.13765* (2025).
- Rohan Surana, Gagan Mundada, Xunyi Jiang, Chuhan Wang, Zhenwei Tang, Difan Jiao, Zihan Huang, Yuxin Xiong, Junda Wu, Sheldon Yu, Xintong Li, Raghav Jain, Nikki Kuang, Sizhe Zhou, Bowen Jin, Zhendong Chu, Tong Yu, Ryan Rossi, Kuan-Hao Huang, Jingbo Shang, Jiawei Han, and Julian McAuley. 2026. Generate, Filter, Control, Replay: A Comprehensive Survey of Rollout Strategies for LLM Reinforcement Learning. (2026). [arXiv:2605.02913](#)
- Richard S Sutton and Andrew G Barto. 1998. *Reinforcement learning: An introduction*. Vol. 1. MIT press Cambridge.
- Mirac Suzgun, Mert Yuksekgonul, Federico Bianchi, Dan Jurafsky, and James Zou. 2026. Dynamic Cheatsheet: Test-Time Learning with Adaptive Memory. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, Vera Demberg, Kentaro Inui, and Lluís Marquez (Eds.). Association for Computational Linguistics, Rabat, Morocco, 7080–7106. [doi:10.18653/v1/2026.eacl-long.333](#)
- Nathan J Szymanski, Bernardus Rendy, Yuxing Fei, Rishi E Kumar, Tanjin He, David Milsted, Matthew J McDermott, Max Gallant, Ekin Dogus Cubuk, and Amil Merchant. 2023. An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature* 624, 7990 (2023), 86.
- Huajie Tan, Sixiang Chen, Yijie Xu, Zixiao Wang, Yuheng Ji, Cheng Chi, Yaoxu Lyu, Zhongxia Zhao, Xiansheng Chen, Peterson Co, Shaoxuan Xie, Guocai Yao, Pengwei Wang, Zhongyuan Wang, and Shanghang Zhang. 2025a. Robo-Dopamine: General Process Reward Modeling for High-Precision Robotic Manipulation. (2025). [arXiv:2512.23703](#)
- Manuscript submitted to ACM

- Haoran Tan, Zeyu Zhang, Chen Ma, Xu Chen, Quanyu Dai, and Zhenhua Dong. 2025b. Membench: Towards more comprehensive evaluation on the memory of llm-based agents. In *Findings of the Association for Computational Linguistics: ACL 2025*. 19336–19352.
- Sijun Tan, Siyuan Zhuang, Kyle Montgomery, William Y. Tang, Alejandro Cuadron, Chenguang Wang, Raluca Ada Popa, and Ion Stoica. 2024. JudgeBench: A Benchmark for Evaluating LLM-based Judges. (2024). arXiv:2410.12784
- Zhengyang Tang, Ziniu Li, Zhenyang Xiao, Tian Ding, Ruoyu Sun, Benyou Wang, Dayiheng Liu, Fei Huang, Tianyu Liu, Bowen Yu, and Junyang Lin. 2025. RealCritic: Towards Effectiveness-Driven Evaluation of Language Model Critiques. (2025). arXiv:2501.14492
- Zhengwei Tao, Ting-En Lin, Xiancai Chen, Hangyu Li, Yuchuan Wu, Yongbin Li, Zhi Jin, Fei Huang, Dacheng Tao, and Jingren Zhou. 2024. A Survey on Self-Evolution of Large Language Models. (2024). arXiv:2404.14387
- Mia Taylor, James Chua, Jan Betley, Johannes Treutlein, and Owain Evans. 2025. School of Reward Hacks: Hacking harmless tasks generalizes to misaligned behavior in LLMs. arXiv:2508.17511
- Nitya Thakkar, Mert Yuksekgonul, Jake Silberg, Animesh Garg, Nanyun Peng, Fei Sha, Rose Yu, Carl Vondrick, and James Zou. 2025. Can LLM feedback enhance review quality? A randomized study of 20k reviews at ICLR 2025. *arXiv preprint arXiv:2504.09737* (2025).
- Jonathan Uesato, Nate Kushman, Ramana Kumar, Francis Song, Noah Siegel, Lisa Wang, Antonia Creswell, Geoffrey Irving, and Irina Higgins. 2022. Solving math word problems with process-and outcome-based feedback. *arXiv preprint arXiv:2211.14275* (2022).
- Peter Vamplew, Benjamin J. Smith, Johan Kallstrom, Gabriel Ramos, Roxana Radulescu, Diederik M. Roijers, Conor F. Hayes, Fredrik Heintz, Patrick Mannion, Pieter J. K. Libin, Richard Dazeley, and Cameron Foale. 2021. Scalar reward is not enough: A response to Silver, Singh, Precup and Sutton (2021). arXiv:2112.15422 [cs.AI] <https://arxiv.org/abs/2112.15422>
- V Venkatesh, Mandeep Rathee, and Avishek Anand. 2025. Trust but Verify! A Survey on Verification Design for Test-time Scaling. (2025). arXiv:2508.16665
- Vijay Viswanathan, Yanchao Sun, Shuang Ma, Xiang Kong, Meng Cao, Graham Neubig, and Tongshuang Wu. 2025. Checklists Are Better Than Reward Models For Aligning Language Models. (2025). arXiv:2507.18624
- Manya Wadhwa, Xinyu Zhao, Junyi Jessy Li, and Greg Durrett. 2024. Learning to refine with fine-grained natural language feedback. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. 12281–12308.
- Eric Wallace, Kai Xiao, Reimar Leike, Lilian Weng, Johannes Heidecke, and Alex Beutel. 2024. The instruction hierarchy: Training llms to prioritize privileged instructions. *arXiv preprint arXiv:2404.13208* (2024).
- Ante Wang, Linfeng Song, Ye Tian, Dian Yu, Haitao Mi, Xiangyu Duan, Zhaopeng Tu, Jinsong Su, and Dong Yu. 2025g. Don't Get Lost in the Trees: Streamlining LLM Reasoning by Overcoming Tree Search Exploration Pitfalls. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (Eds.). Association for Computational Linguistics, Vienna, Austria, 23946–23959. doi:10.18653/v1/2025.acl-long.1167
- Chenglong Wang, Yongyu Mu, Hang Zhou, Yifu Huo, Ziming Zhu, Jiali Zeng, Murun Yang, Bei Li, Xiaoyang Hao, Chunliang Zhang, Fandong Meng, Jingbo Zhu, and Tong Xiao. 2025f. GRAM-R²: Self-Training Generative Foundation Reward Models for Reward Reasoning. (2025). arXiv:2509.02492
- Guoqing Wang, Sunhao Dai, Guangze Ye, Zeyu Gan, Wei Yao, Yong Deng, Xiaofeng Wu, and Zhenzhe Ying. 2025a. Information Gain-based Policy Optimization: A Simple and Effective Approach for Multi-Turn Search Agents. (2025). arXiv:2510.14967
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2023c. Voyager: An Open-Ended Embodied Agent with Large Language Models. *arXiv preprint arXiv:2305.16291* (2023).
- Hanyang Wang, Lu Wang, Chaoyun Zhang, Tianjun Mao, Si Qin, Qingwei Lin, Saravan Rajmohan, and Dongmei Zhang. 2025h. Text2Grad: Reinforcement Learning from Natural Language Feedback. *arXiv preprint arXiv:2505.22338* (2025).
- Jun Wang. 2025. Memento 2: Learning by Stateful Reflective Memory. *arXiv preprint arXiv:2512.22716* (2025). arXiv:2512.22716
- Jiaxuan Wang, Yulan Hu, Wenjin Yang, Zheng Pan, Xin Li, and Lan-Zhe Guo. 2026b. Aligning Agents via Planning: A Benchmark for Trajectory-Level Reward Modeling. (2026). arXiv:2604.08178
- Jianyu Wang, Zhiqiang Hu, and Lidong Bing. 2025c. Evolving Prompts In-Context: An Open-ended, Self-replicating Perspective. (2025). arXiv:2506.17930
- Peiyi Wang, Lei Li, Zhihong Shao, RX Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. 2024d. Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations. *arXiv preprint arXiv:2312.08935* (2024).
- Pengkai Wang, Pengwei Liu, Qi Zuo, Zhijie Sang, Congkai Xie, and Hongxia Yang. 2025e. InfMed-ORBIT: Aligning LLMs on Open-Ended Complex Tasks via Rubric-Based Incremental Training. (2025). arXiv:2510.15859
- Qibin Wang, Pu Zhao, Shaohan Huang, Fangkai Yang, Lu Wang, Furu Wei, Qingwei Lin, Saravan Rajmohan, and Dongmei Zhang. 2025j. Learning to refine: Self-refinement of parallel reasoning in llms. *arXiv preprint arXiv:2509.00084* (2025).
- Shuhe Wang, Shengyu Zhang, Jie Zhang, Runyi Hu, Xiaoya Li, Tianwei Zhang, Jiwei Li, Fei Wu, Guoyin Wang, and Eduard Hovy. 2024g. Reinforcement Learning Enhanced LLMs: A Survey. (2024). arXiv:2412.10400
- Tianlu Wang, Ilia Kulikov, Olga Golovneva, Ping Yu, Weizhe Yuan, Jane Dwivedi-Yu, Richard Yuanzhe Pang, Maryam Fazel-Zarandi, Jason Weston, and Xian Li. 2024c. Self-Taught Evaluators. (2024). arXiv:2408.02666
- Tianlu Wang, Ping Yu, Xiaoqing Ellen Tan, Sean O'Brien, Ramakanth Pasunuru, Jane Dwivedi-Yu, Olga Golovneva, Luke Zettlemoyer, Maryam Fazel-Zarandi, and Asli Celikyilmaz. 2023d. Shepherd: A Critic for Language Model Generation. *arXiv preprint arXiv:2308.04592* (2023).
- Wenyi Wang, Hisham A. Alyahya, Dylan R. Ashley, Oleg Serikov, Dmitrii Khizbullin, Francesco Faccio, and Jürgen Schmidhuber. 2024a. How to Correctly do Semantic Backpropagation on Language-based Agentic Systems. (2024). arXiv:2412.03624
- Xingyao Wang, Yangyi Chen, Lifan Yuan, Yizhe Zhang, Yunzhu Li, Hao Peng, and Heng Ji. 2024b. Executable code actions elicit better llm agents. In *Forty-first International Conference on Machine Learning*.

- Xuekang Wang, Zhuoyuan Hao, Shuo Hou, Hao Peng, Juanzi Li, and Xiaozhi Wang. 2026a. Reproducing, Analyzing, and Detecting Reward Hacking in Rubric-Based Reinforcement Learning. [arXiv:2606.04923](#)
- Xingyao Wang, Boxuan Li, Yufan Song, Frank F Xu, Xiangru Tang, Mingchen Zhuge, Jiayi Pan, Yueqi Song, Bowen Li, and Jaskirat Singh. 2025d. Openhands: An open platform for ai software developers as generalist agents. In *International Conference on Learning Representations*, Vol. 2025. 65882–65919.
- Xingyao Wang, Zihan Wang, Jiateng Liu, Yangyi Chen, Lifan Yuan, Hao Peng, and Heng Ji. 2023b. MINT: Evaluating LLMs in Multi-turn Interaction with Tools and Language Feedback. (2023). [arXiv:2309.10691](#)
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2022. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171* (2022).
- Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A Smith, Daniel Khashabi, and Hannaneh Hajishirzi. 2023a. Self-instruct: Aligning language models with self-generated instructions. In *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers)*. 13484–13508.
- Yufei Wang, Zhanyi Sun, Jesse Zhang, Zhou Xian, Erdem Biyik, David Held, and Zackory Erickson. 2024f. RL-VLM-F: Reinforcement Learning from Vision Language Foundation Model Feedback. (2024). [arXiv:2402.03681](#)
- Yubo Wang, Xiang Yue, and Wenhui Chen. 2025i. Critique Fine-Tuning: Learning to Critique is More Effective than Learning to Imitate. *arXiv preprint* (2025).
- Zongqi Wang, Tianle Gu, Chen Gong, Xin Tian, Siqi Bao, and Yujiu Yang. 2025b. SCAN: Structured Capability Assessment and Navigation for LLMs. (2025). [arXiv:2505.06698](#)
- Zongqi Wang, Rui Wang, Yuchuan Wu, Yiyao Yu, Pinyi Zhang, Shaoning Sun, Yujiu Yang, and Yongbin Li. 2026c. Reward Modeling from Natural Language Human Feedback. (2026). [arXiv:2601.07349](#)
- Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. 2024e. Agent Workflow Memory. (2024). [arXiv:2409.07429](#)
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* 35 (2022), 24824–24837.
- Jiaqi Wei, Xiang Zhang, Yuejin Yang, Wenxuan Huang, Juntai Cao, Sheng Xu, Xiang Zhuang, Zhangyang Gao, Muhammad Abdul-Mageed, Laks V. S. Lakshmanan, Chenyu You, Wanli Ouyang, and Siqi Sun. 2025c. Unifying Tree Search Algorithm and Reward Design for LLM Reasoning: A Survey. (2025). [arXiv:2510.09988](#)
- Quan Wei, Siliang Zeng, Chenliang Li, William Brown, Oana Frunza, Wei Deng, Anderson Schneider, Yuriy Nevmyvaka, Yang Katie Zhao, Alfredo Garcia, and Mingyi Hong. 2025b. Reinforcing Multi-Turn Reasoning in LLM Agents via Turn-Level Reward Design. (2025). [arXiv:2505.11821](#)
- Tianxin Wei, Naveen Sachdeva, Benjamin Coleman, Zhankui He, Yuanchen Bei, Xuying Ning, Mengting Ai, Yunzhe Li, Jingrui He, Ed H. Chi, Chi Wang, Shuo Chen, Fernando Pereira, Wang-Cheng Kang, and Derek Zhiyuan Cheng. 2025a. Evo-Memory: Benchmarking LLM Agent Test-time Learning with Self-Evolving Memory. (2025). [arXiv:2511.20857](#)
- Sean Welleck, Ximing Lu, Peter West, Faeze Brahman, Tianxiao Shen, Daniel Khashabi, and Yejin Choi. 2023. Generating Sequences by Learning to Self-Correct. In *International Conference on Learning Representations*. [arXiv:2211.00053](#)
- Bosi Wen, Yilin Niu, Cunxiang Wang, Xiaoying Ling, Ying Zhang, Pei Ke, Hongning Wang, and Minlie Huang. 2026. IF-RewardBench: Benchmarking Judge Models for Instruction-Following Evaluation. (2026). [arXiv:2603.04738](#)
- Yixuan Weng, Minjun Zhu, Fei Xia, Bin Li, Shizhu He, Shengping Liu, Bin Sun, Kang Liu, and Jun Zhao. 2023. Large language models are better reasoners with self-verification. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 2550–2575.
- Chelsea C White III and Douglas J White. 1989. Markov decision processes. *European Journal of Operational Research* 39, 1 (1989), 1–16.
- Chenxi Whitehouse, Tianlu Wang, Ping Yu, Xian Li, Jason Weston, Ilia Kulikov, and Swarnadeep Saha. 2025. J1: Incentivizing Thinking in LLM-as-a-Judge via Reinforcement Learning. (2025). [arXiv:2505.10320](#)
- Fang Wu, Aaron Tu, Weihao Xuan, Heli Qi, Xu Huang, Qingcheng Zeng, Shayan Talaei, Yijia Xiao, Peng Xia, Xiangru Tang, Yuchen Zhuang, Yinxin Li, Bing Hu, Hanqun Cao, Wenqi Shi, Rui Yang, Nan Liu, Huaxiu Yao, Ge Liu, Li Erran Li, Amin Saberi, Naoto Yokoya, Jure Leskovec, and Yejin Choi. 2025a. Position: The Hidden Costs and Measurement Gaps of Reinforcement Learning with Verifiable Rewards. (2025). [arXiv:2509.21882](#)
- Mian Wu, Gavin Zhang, Sewon Min, Sergey Levine, and Aviral Kumar. 2025b. RLAC: Reinforcement Learning with Adversarial Critic for Free-Form Generation Tasks. (2025). [arXiv:2511.01758](#)
- Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryan W White, Doug Burger, and Chi Wang. 2023. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation Framework. *arXiv preprint arXiv:2308.08155* (2023).
- Shengguang Wu, Hao Zhu, Yuhui Zhang, Xiaohan Wang, and Serena Yeung-Levy. 2026b. AutoMem: Automated Learning of Memory as a Cognitive Skill. (2026). [arXiv:2607.01224](#)
- Tianhao Wu, Weizhe Yuan, Olga Golovneva, Jing Xu, Yuandong Tian, Jiantao Jiao, Jason Weston, and Sainbayar Sukhbaatar. 2024. Meta-Rewarding Language Models: Self-Improving Alignment with LLM-as-a-Meta-Judge. (2024). [arXiv:2407.19594](#)
- Xiaobao Wu. 2025. Sailing by the Stars: A Survey on Reward Models and Learning Strategies for Learning from Rewards. (2025). [arXiv:2505.02686](#)
- Yanru Wu, Weiduo Yuan, Ang Qi, Vitor Guizilini, Jiageng Mao, and Yue Wang. 2026a. Large Reward Models: Generalizable Online Robot Reward Generation with Vision-Language Models. (2026). [arXiv:2603.16065](#)
- Manuscript submitted to ACM

- Evžen Wybitul, Evan Ryan Gunter, Mikhail Seleznyov, and David Lindner. 2024. ViSta Dataset: Do vision-language models understand sequential tasks? (2024). [arXiv:2411.13211](#)
- Jiajun Xi, Yinong He, Jianing Yang, Yinpei Dai, and Joyce Chai. 2024a. Teaching Embodied Reinforcement Learning Agents: Informativeness and Diversity of Language Use. (2024). [arXiv:2410.24218](#)
- Zhiheng Xi, Jixuan Huang, Xin Guo, Boyang Hong, Dingwen Yang, Xiaoran Fan, Shuo Li, Zehui Chen, Junjie Ye, Siyu Yuan, Zhengyin Du, Xuesong Yao, Yufei Xu, Jiecao Chen, Rui Zheng, Tao Gui, Qi Zhang, and Xuanjing Huang. 2025. Critique-RL: Training Language Models for Critiquing through Two-Stage Reinforcement Learning. (2025). [arXiv:2510.24320](#)
- Zhiheng Xi, Dingwen Yang, Jixuan Huang, Jiafu Tang, Guanyu Li, Yiwen Ding, Wei He, Boyang Hong, Shihan Do, Wenyu Zhan, Xiao Wang, Rui Zheng, Tao Ji, Xiaowei Shi, Yitao Zhai, Rongxiang Weng, Jingang Wang, Xunliang Cai, Tao Gui, Zuxuan Wu, Qi Zhang, Xipeng Qiu, Xuanjing Huang, and Yu-Gang Jiang. 2024b. Enhancing LLM Reasoning via Critique Models with Test-Time and Training-Time Supervision. (2024). [arXiv:2411.16579](#)
- Peng Xia, Jianwen Chen, Hanyang Wang, Jiaqi Liu, Kaide Zeng, Yu Wang, Siwei Han, Yiyang Zhou, Xujiang Zhao, Haifeng Chen, Zeyu Zheng, Cihang Xie, and Huaxiu Yao. 2026. SkillRL: Evolving Agents via Recursive Skill-Augmented Reinforcement Learning. (2026). [arXiv:2602.08234](#)
- Jinyu Xiang, Jiayi Zhang, Zhaoyang Yu, Xinbing Liang, Fengwei Teng, Jinhao Tu, Fashen Ren, Xiangru Tang, Sirui Hong, Chenglin Wu, and Yuyu Luo. 2025. Self-Supervised Prompt Optimization. (2025). [arXiv:2502.06855](#)
- Tim Z. Xiao, Robert Bamler, Bernhard Schölkopf, and Weiyang Liu. 2024a. Verbalized Machine Learning: Revisiting Machine Learning with Language Models. (2024). [arXiv:2406.04344](#)
- Wenyi Xiao, Zechuan Wang, Leilei Gan, Shuai Zhao, Zongrui Li, Ruirui Lei, Wanggui He, Luu Anh Tuan, Long Chen, Hao Jiang, Zhou Zhao, and Fei Wu. 2024b. A Comprehensive Survey of Direct Preference Optimization: Datasets, Theories, Variants, and Applications. (2024). [arXiv:2410.15595](#)
- Guofu Xie, Yunsheng Shi, Hongtao Tian, Ting Yao, and Xiao Zhang. 2025b. CAPO: Towards Enhancing LLM Reasoning through Generative Credit Assignment. (2025). [arXiv:2508.02298](#)
- Tianbao Xie, Siheng Zhao, Chen Wu, Yitao Liu, Qian Luo, Victor Zhong, Yanchao Yang, and Tao Yu. 2024. Text2reward: Reward shaping with language models for reinforcement learning. In *International Conference on Learning Representations*, Vol. 2024. 35663–35699.
- Yuxi Xie, Kenji Kawaguchi, Yiran Zhao, James Xu Zhao, Min-Yen Kan, Junxian He, and Michael Xie. 2023. Self-evaluation guided beam search for reasoning. *Advances in Neural Information Processing Systems* 36 (2023), 41618–41650.
- Zhihui Xie, Jie Chen, Liyu Chen, Weichao Mao, Jingjing Xu, and Lingpeng Kong. 2025a. Teaching Language Models to Critique via Reinforcement Learning. (2025). [arXiv:2502.03492](#)
- Wei Xiong, Wenting Zhao, Weizhe Yuan, Olga Golovneva, Tong Zhang, Jason Weston, and Sainbayar Sukhbaatar. 2025. StepWiser: Stepwise Generative Judges for Wiser Reasoning. (2025). [arXiv:2508.19229](#)
- Fengli Xu, Qian Yue Hao, Zefang Zong, Jingwei Wang, Yunke Zhang, Jingyi Wang, Xiaochong Lan, Jiahui Gong, Tianjian Ouyang, Fanjin Meng, Chenyang Shao, Yuwei Yan, Qinglong Yang, Yiwen Song, Sijian Ren, Xinyuan Hu, Yu Li, Jie Feng, Chen Gao, and Yong Li. 2025a. Towards Large Reasoning Models: A Survey of Reinforced Reasoning with Large Language Models. (2025). [arXiv:2501.09686](#)
- Guowei Xu, Mert Yuksekgonul, Carlos Guestrin, and James Zou. 2025d. metaTextGrad: Automatically Optimizing Language Model Optimizers. (2025). [arXiv:2505.18524](#)
- Tianze Xu, Yanzhao Zheng, Pengrui Lu, Lyumanshan Ye, Yong Wu, Zhentao Zhang, Yuanqiang Yu, Chao Ma, Jihuai Zhu, Pengfei Liu, Baohua Dong, Hangcheng Zhu, Ruohui Huang, and Gang Yu. 2026. Rubrics to Tokens: Bridging Response-level Rubrics and Token-level Rewards in Instruction Following Tasks. (2026). [arXiv:2604.02795](#)
- Wenda Xu, Daniel Deutsch, Mara Finkelstein, Juraj Juraska, Biao Zhang, Zhongtao Liu, William Yang Wang, Lei Li, and Markus Freitag. 2024. LLMRefine: Pinpointing and Refining Large Language Models via Fine-Grained Actionable Feedback. *arXiv preprint arXiv:2311.09336* (2024).
- Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025b. A-MEM: Agentic Memory for LLM Agents. (2025). [arXiv:2502.12110](#)
- Wanqiao Xu, Allen Nie, Ruijie Zheng, Aditya Modi, Adith Swaminathan, and Ching-An Cheng. 2025c. Formalizing Learning from Language Feedback with Provable Guarantees. (2025). [arXiv:2506.10341](#)
- Sikuan Yan, Ahmed Bahloul, Ercong Nie, Susanna Schwarzmann, Riccardo Trivisonno, Volker Tresp, and Yunpu Ma. 2026. Memory-R2: Fair Credit Assignment for Long-Horizon Memory-Augmented LLM Agents. (2026). [arXiv:2605.21768](#)
- Sikuan Yan, Xiufeng Yang, Zuchao Huang, Ercong Nie, Zifeng Ding, Zonggen Li, Xiaowen Ma, Jinhe Bi, Kristian Kersting, Jeff Z. Pan, Hinrich Schütze, Volker Tresp, and Yunpu Ma. 2025b. Memory-R1: Enhancing Large Language Model Agents to Manage and Utilize Memories via Reinforcement Learning. (2025). [arXiv:2508.19828](#)
- Yuchen Yan, Jin Jiang, Zhenbang Ren, Yijun Li, Xudong Cai, Yang Liu, Xin Xu, Mengdi Zhang, Jian Shao, Yongliang Shen, Jun Xiao, and Yueting Zhuang. 2025a. VerifyBench: Benchmarking Reference-based Reward Systems for Large Language Models. (2025). [arXiv:2505.15801](#)
- Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V Le, Denny Zhou, and Xinyun Chen. 2024b. Large Language Models as Optimizers. In *International Conference on Learning Representations Workshop*. [arXiv:2309.03409](#)
- Cheng Yang, Xueming Yang, Licheng Wen, Daocheng Fu, Jianbiao Mei, Rong Wu, Pinlong Cai, Yufan Shen, Nianchen Deng, Botian Shi, Yu Qiao, and Haifeng Li. 2025b. Learning on the Job: An Experience-Driven Self-Evolving Agent for Long-Horizon Tasks. (2025). [arXiv:2510.08002](#)
- John Yang, Carlos E Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. 2024a. Swe-agent: Agent-computer interfaces enable automated software engineering. *Advances in Neural Information Processing Systems* 37 (2024), 50528–50652.
- John Yang, Akshara Prabhakar, Karthik Narasimhan, and Shunyu Yao. 2023. Intercode: Standardizing and benchmarking interactive coding with execution feedback. *Advances in Neural Information Processing Systems* 36 (2023), 23826–23854.

- Ruihan Yang, Fanghua Ye, Jian Li, Siyu Yuan, Yikai Zhang, Zhaopeng Tu, Xiaolong Li, and Deqing Yang. 2025c. The Lighthouse of Language: Enhancing LLM Agents via Critique-Guided Improvement. (2025). [arXiv:2503.16024](#)
- Wenkai Yang, Jingwen Chen, Yankai Lin, and Ji-Rong Wen. 2025a. DeepCritic: Deliberate Critique with Large Language Models. (2025). [arXiv:2505.00662](#)
- Xianglin Yang, Bryan Hooi, Gelei Deng, Tianwei Zhang, and Jin Song Dong. 2026a. Turning Bias into Bugs: Bandit-Guided Style Manipulation Attacks on LLM Judges. [arXiv:2605.26156](#)
- Yutao Yang, Junsong Li, Qianjun Pan, Bihao Zhan, Yuxuan Cai, Lin Du, Jie Zhou, Kai Chen, Qin Chen, Xin Li, Bo Zhang, and Liang He. 2026b. AutoSkill: Experience-Driven Lifelong Learning via Skill Self-Evolution. (2026). [arXiv:2603.01145](#)
- Yongjin Yang, Euiin Yi, Jongwoo Ko, Kimin Lee, Zhijing Jin, and Se-Young Yun. 2025d. Revisiting Multi-Agent Debate as Test-Time Scaling: A Systematic Study of Conditional Effectiveness. (2025). [arXiv:2505.22960](#)
- Shunyu Yao, Rohan Rao, Matthew Hausknecht, and Karthik Narasimhan. 2020. Keep calm and explore: Language models for action generation in text-based games. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 8736–8754.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. 2023a. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems* 36 (2023), 11809–11822.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023b. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations*.
- Jiayi Ye, Yanbo Wang, Yue Huang, Dongping Chen, Qihui Zhang, Nuno Moniz, Tian Gao, Werner Geyer, Chao Huang, Pin-Yu Chen, Nitesh V Chawla, and Xiangliang Zhang. 2024. Justice or Prejudice? Quantifying Biases in LLM-as-a-Judge. (2024). [arXiv:2410.02736](#)
- Zhilong Ye, Yun Yue, Haowen Wang, Xudong Han, Jiadi Jiang, Cheng Wei, Lei Fan, Jiaxin Liang, Shuowen Zhang, Ji Li, Chunxiao Guo, Jian Wang, Peng Wei, and Jinjie Gu. 2025. Self-Rewarding Rubric-Based Reinforcement Learning for Open-Ended Reasoning. (2025). [arXiv:2509.25534](#)
- John Seon Keun Yi, Aaron Mueller, and Dokyun Lee. 2026. Latent Agents: A Post-Training Procedure for Internalized Multi-Agent Debate. (2026). [arXiv:2604.24881](#)
- Li Yin and Zhangyang Wang. 2025. LLM-AutoDiff: Auto-Differentiate Any LLM Workflow. (2025). [arXiv:2501.16673](#)
- Zhangyue Yin, Qiushi Sun, Zhiyuan Zeng, Qinyuan Cheng, Xipeng Qiu, and Xuanjing Huang. 2025. Dynamic and Generalizable Process Reward Modeling. (2025). [arXiv:2507.17849](#)
- Runyang You, Hongru Cai, Caiqi Zhang, Qiancheng Xu, Meng Liu, Tiezheng Yu, Yongqi Li, and Wenjie Li. 2026. Agent-as-a-Judge. (2026). [arXiv:2601.05111](#)
- Wenhao Yu, Nimrod Gileadi, Chuyuan Fu, Sean Kirmani, Kuang-Huei Lee, Montse Gonzalez Arenas, Hao-Tien Lewis Chiang, Tom Erez, Leonard Hasenclever, and Jan Humplik. 2023. Language to rewards for robotic skill synthesis. *arXiv preprint arXiv:2306.08647* (2023).
- Xiao Yu, Baolin Peng, Michel Galley, Jianfeng Gao, and Zhou Yu. 2024b. Teaching Language Models to Self-Improve through Interactive Demonstrations. In *North American Chapter of the Association for Computational Linguistics*.
- Yue Yu, Zhengxing Chen, Aston Zhang, Liang Tan, Chenguang Zhu, Richard Yuanzhe Pang, Yundi Qian, Xuwei Wang, Suchin Gururangan, Chao Zhang, Melanie Kambadur, Dhruv Mahajan, and Rui Hou. 2024a. Self-Generated Critiques Boost Reward Modeling for Language Models. (2024). [arXiv:2411.16646](#)
- Siyu Yuan, Zehui Chen, Zhiheng Xi, Junjie Ye, Zhengyin Du, and Jiecao Chen. 2025a. Agent-R: Training Language Model Agents to Reflect via Iterative Self-Training. (2025). [arXiv:2501.11425](#)
- Weizhe Yuan, Richard Yuanzhe Pang, Kyunghyun Cho, Xian Li, Sainbayar Sukhbaatar, Jing Xu, and Jason Weston. 2024. Self-rewarding language models. *arXiv preprint arXiv:2401.10020* (2024).
- Youliang Yuan, Qiuyang Mang, Jingbang Chen, Hong Wan, Xiaoyuan Liu, Junlielong Xu, Jen-tse Huang, Wenxuan Wang, Wenxiang Jiao, and Pinjia He. 2025b. Curing Miracle Steps in LLM Mathematical Reasoning with Rubric Rewards. (2025). [arXiv:2510.07774](#)
- Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. 2025. Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model? *arXiv preprint arXiv:2504.13837* (2025). [arXiv:2504.13837](#)
- Mert Yuksekgonul, Federico Bianchi, Joseph Boen, Sheng Liu, Zhi Huang, Carlos Guestrin, and James Zou. 2024. TextGrad: Automatic “Differentiation” via Text. *arXiv preprint arXiv:2406.07496* (2024).
- Mert Yuksekgonul, Federico Bianchi, Joseph Boen, Sheng Liu, Pan Lu, Zhi Huang, Carlos Guestrin, and James Zou. 2025. Optimizing generative AI by backpropagating language model feedback. *Nature* (2025).
- Eric Zelikman, Georges Harik, Yijia Shao, Varuna Jayasiri, Nick Haber, and Noah D Goodman. 2024. Quiet-STaR: Language Models Can Teach Themselves to Think Before Speaking. *arXiv preprint arXiv:2403.09629* (2024).
- Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah D Goodman. 2022. STaR: Bootstrapping Reasoning With Reasoning. In *Advances in Neural Information Processing Systems*.
- Runhao Zeng, Dingjie Zhou, Qiwei Liang, Junlin Liu, Hui Li, Changxin Huang, Jianqiang Li, Xiping Hu, and Fuchun Sun. 2024. Video2reward: Generating reward function from videos for legged robot behavior learning. *arXiv preprint arXiv:2412.05515* (2024).
- Zhiyuan Zeng, Jiatong Yu, Tianyu Gao, Yu Meng, Tanya Goyal, and Danqi Chen. 2023. Evaluating Large Language Models at Evaluating Instruction Following. (2023). [arXiv:2310.07641](#)
- Lihan Zha, Yuchen Cui, Li-Heng Lin, Minae Kwon, Montserrat Gonzalez Arenas, Andy Zeng, Fei Xia, and Dorsa Sadigh. 2023. Distilling and Retrieving Generalizable Knowledge for Robot Manipulation via Language Corrections. (2023). [arXiv:2311.10678](#)
- Qiusi Zhan, Richard Fang, Henil Shalin Panchal, and Daniel Kang. 2025. Adaptive attacks break defenses against indirect prompt injection attacks on llm agents. In *Findings of the Association for Computational Linguistics: NAACL 2025*. 7101–7117.
- Manuscript submitted to ACM

- Qiusi Zhan, Zhixiang Liang, Zifan Ying, and Daniel Kang. 2024. Injecagent: Benchmarking indirect prompt injections in tool-integrated large language model agents. In *Findings of the Association for Computational Linguistics: ACL 2024*. 10471–10506.
- Alexander Zhang, Marcus Dong, Jiaheng Liu, Wei Zhang, Yejie Wang, Jian Yang, Ge Zhang, Tianyu Liu, Zhongyuan Peng, Yingshui Tan, Yuanxing Zhang, Zhexu Wang, Weixun Wang, Yancheng He, Ken Deng, Wangchunshu Zhou, Wenhao Huang, and Zhaoxiang Zhang. 2025d. CodeCriticBench: A Holistic Code Critique Benchmark for Large Language Models. (2025). arXiv:2502.16614
- Chenchen Zhang. 2026. From Reasoning to Agentic: Credit Assignment in Reinforcement Learning for Large Language Models. (2026). arXiv:2604.09459
- Dylan Zhang, Yanshan Lin, Zhengkun Wu, Yihang Sun, Bingxuan Li, Dianqi Li, and Hao Peng. 2026b. Useful Memories Become Faulty When Continuously Updated by LLMs. (2026). arXiv:2605.12978
- Guibin Zhang, Muxin Fu, Guancheng Wan, Miao Yu, Kun Wang, and Shuicheng Yan. 2025e. G-Memory: Tracing Hierarchical Memory for Multi-Agent Systems. (2025). arXiv:2506.07398
- Guibin Zhang, Hejia Geng, Xiaohang Yu, Zhenfei Yin, Zaibin Zhang, Zelin Tan, Heng Zhou, Zhongzhi Li, Xiangyuan Xue, Yijiang Li, Yifan Zhou, Yang Chen, Chen Zhang, Yutao Fan, Zihu Wang, Songtao Huang, Francisco Piedrahita-Velez, Yue Liao, Hongru Wang, Mengyue Yang, Heng Ji, Jun Wang, Shuicheng Yan, Philip Torr, and Lei Bai. 2025f. The Landscape of Agentic Reinforcement Learning for LLMs: A Survey. (2025). arXiv:2509.02547
- Guibin Zhang, Haotian Ren, Chong Zhan, Zhenhong Zhou, Junhao Wang, He Zhu, Wangchunshu Zhou, and Shuicheng Yan. 2025j. MemEvolve: Meta-Evolution of Agent Memory Systems. (2025). arXiv:2512.18746
- Hangfan Zhang, Zhiyao Cui, Jianhao Chen, Xinrun Wang, Qiaosheng Zhang, Zhen Wang, Dinghao Wu, and Shuyue Hu. 2025a. Stop Overvaluing Multi-Agent Debate – We Must Rethink Evaluation and Embrace Model Heterogeneity. (2025). arXiv:2502.08788
- Haozhen Zhang, Quanyu Long, Jianzhu Bao, Tao Feng, Weizhi Zhang, Haodong Yue, and Wenya Wang. 2026c. MemSkill: Learning and Evolving Memory Skills for Self-Evolving Agents. (2026). arXiv:2602.02474
- Jiayi Zhang, Jinyu Xiang, Zhaoyang Yu, Fengwei Teng, Xionghui Chen, Jiaqi Chen, Mingchen Zhuge, Xin Cheng, Sirui Hong, Jinlin Wang, Bingnan Zheng, Bang Liu, Yuyu Luo, and Chenglin Wu. 2024b. AFlow: Automating Agentic Workflow Generation. (2024). arXiv:2410.10762
- Kaiyan Zhang, Yuxin Zuo, Bingxiang He, Youbang Sun, Runze Liu, Che Jiang, Yuchen Fan, Kai Tian, Guoli Jia, Pengfei Li, Yu Fu, Xingtai Lv, Yuchen Zhang, Sihang Zeng, Shang Qu, Haozhan Li, Shijie Wang, Yuru Wang, Xinwei Long, Fangfu Liu, Xiang Xu, Jiaze Ma, Xuekai Zhu, Ermo Hua, Yihao Liu, Zonglin Li, Huayu Chen, Xiaoye Qu, Yafu Li, Weize Chen, Zhenzhao Yuan, Junqi Gao, Dong Li, Zhiyuan Ma, Ganqu Cui, Zhiyuan Liu, Biqing Qi, Ning Ding, and Bowen Zhou. 2025m. A Survey of Reinforcement Learning for Large Reasoning Models. (2025). arXiv:2509.08827
- Lunjun Zhang, Ryan Chen, and Bradley C. Stadie. 2026a. Evolutionary System Prompt Learning for Reinforcement Learning in LLMs. (2026). arXiv:2602.14697
- Lunjun Zhang, Arian Hosseini, Hritik Bansal, Mehran Kazemi, Aviral Kumar, and Rishabh Agarwal. 2025g. Generative Verifiers: Reward Modeling as Next-Token Prediction. In *International Conference on Learning Representations*. arXiv:2408.15240
- Peiyan Zhang, Haibo Jin, Leyang Hu, Xinnuo Li, Liying Kang, Man Luo, Yangqiu Song, and Haohan Wang. 2024a. REVOLVE: Optimizing AI Systems by Tracking Response Evolution in Textual Optimization. (2024). arXiv:2412.03092
- Qizheng Zhang, Changran Hu, Shubhangi Upasani, Boyuan Ma, Fenglu Hong, Vamsidhar Kamanuru, Jay Rainton, Chen Wu, Mengmeng Ji, Hanchen Li, Urmish Thakker, James Zou, and Kunle Olukotun. 2025h. Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models. (2025). arXiv:2510.04618
- Qiyuan Zhang, Fuyuan Lyu, Zexu Sun, Lei Wang, Weixu Zhang, Wenyue Hua, Haolun Wu, Zhihan Guo, Yufei Wang, Niklas Muennighoff, Irwin King, Xue Liu, and Chen Ma. 2025i. A Survey on Test-Time Scaling in Large Language Models: What, How, Where, and How Well? (2025). arXiv:2503.24235
- Xiaohan Zhang, Yan Ding, Yohei Hayamizu, Zainab Altaweel, Yifeng Zhu, Yuke Zhu, Peter Stone, Chris Paxton, and Shiqi Zhang. 2025c. LLM-GROP: Visually grounded robot task and motion planning with large language models. *The International Journal of Robotics Research* (2025), 02783649251378196.
- Yaolun Zhang, Yiran Wu, Yijiong Yu, Qingyun Wu, and Huazheng Wang. 2026e. Live-Evo: Online Evolution of Agentic Memory from Continuous Feedback. (2026). arXiv:2602.02369
- Yang Zhang, Shixin Yang, Chenjia Bai, Fei Wu, Xiu Li, Zhen Wang, and Xuelong Li. 2025k. Towards efficient llm grounding for embodied multi-agent collaboration. In *Findings of the Association for Computational Linguistics: ACL 2025*. 1663–1699.
- Zeyu Zhang, Quanyu Dai, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. 2025b. A survey on the memory mechanism of large language model-based agents. *ACM Transactions on Information Systems* 43, 6 (2025), 1–47.
- Zhang Zhang, Shuqi Lu, Hongjin Qian, Di He, and Zheng Liu. 2026d. AgentFactory: A Self-Evolving Framework Through Executable Subagent Accumulation and Reuse. (2026). arXiv:2603.18000
- Zhenru Zhang, Chujie Zheng, Yangzhen Wu, Beichen Zhang, Runji Lin, Bowen Yu, Dayiheng Liu, Jingren Zhou, and Junyang Lin. 2025l. The Lessons of Developing Process Reward Models in Mathematical Reasoning. (2025). arXiv:2501.07301
- Andrew Zhao, Reshmi Ghosh, Vitor Carvalho, Emily Lawton, Keegan Hines, Gao Huang, and Jack W. Stokes. 2025a. Are My Optimized Prompts Compromised? Exploring Vulnerabilities of LLM-based Optimizers. arXiv:2510.14381
- Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. 2024. ExpeL: LLM Agents Are Experiential Learners. In *AAAI Conference on Artificial Intelligence*.
- Jian Zhao, Runze Liu, Kaiyan Zhang, Zhimu Zhou, Junqi Gao, Dong Li, Jiafei Lyu, Zhouyi Qian, Biqing Qi, and Xiu Li. 2026. Genprm: Scaling test-time compute of process reward models via generative reasoning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 34932–34940.
- Yulai Zhao, Haolin Liu, Dian Yu, Sunyuan Kung, Meijia Chen, Haitao Mi, and Dong Yu. 2025b. One Token to Fool LLM-as-a-Judge. arXiv:2507.08794
- Chujie Zheng, Zhenru Zhang, Beichen Zhang, Runji Lin, Keming Lu, Bowen Yu, Dayiheng Liu, Jingren Zhou, and Junyang Lin. 2024. ProcessBench: Identifying Process Errors in Mathematical Reasoning. (2024). arXiv:2412.06559

- Junhao Zheng, Chengming Shi, Xidi Cai, Qiuke Li, Duzhen Zhang, Chenxing Li, Dong Yu, and Qianli Ma. 2025. Lifelong Learning of Large Language Model based Agents: A Roadmap. (2025). [arXiv:2501.07278](#)
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, and Eric Xing. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems* 36 (2023), 46595–46623.
- Victor Zhong, Dipendra Misra, Xingdi Yuan, and Marc-Alexandre Côté. 2024. Policy Improvement using Language Feedback Models. (2024). [arXiv:2402.07876](#)
- Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. 2024b. Language Agent Tree Search Unifies Reasoning, Acting, and Planning in Language Models. In *International Conference on Machine Learning*.
- Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, and Quoc Le. 2022. Least-to-most prompting enables complex reasoning in large language models. *arXiv preprint arXiv:2205.10625* (2022).
- Enshen Zhou, Qi Su, Cheng Chi, Zhizheng Zhang, Zhongyuan Wang, Tiejun Huang, Lu Sheng, and He Wang. 2025c. Code-as-monitor: Constraint-aware visual programming for reactive and proactive robotic failure detection. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 6919–6929.
- Enyu Zhou, Guodong Zheng, Binghai Wang, Zhiheng Xi, Shihan Dou, Rong Bao, Wei Shen, Limao Xiong, Jessica Fan, Yurong Mou, Rui Zheng, Tao Gui, Qi Zhang, and Xuanjing Huang. 2024e. RMB: Comprehensively Benchmarking Reward Models in LLM Alignment. (2024). [arXiv:2410.09893](#)
- Huichi Zhou, Yihang Chen, Siyuan Guo, Xue Yan, Kin Hei Lee, Zihan Wang, Ka Yiu Lee, Guchun Zhang, Kun Shao, Linyi Yang, and Jun Wang. 2025a. Memento: Fine-tuning LLM Agents without Fine-tuning LLMs. (2025). [arXiv:2508.16153](#)
- Huichi Zhou, Siyuan Guo, Anjie Liu, Zhongwei Yu, Ziqin Gong, Bowen Zhao, Zhixun Chen, Menglong Zhang, Yihang Chen, Jinsong Li, Runyu Yang, Qiangbin Liu, Xinlei Yu, Jianmin Zhou, Na Wang, Chunyang Sun, and Jun Wang. 2026. Memento-Skills: Let Agents Design Agents. (2026). [arXiv:2603.18743](#)
- Han Zhou, Xingchen Wan, Ruoxi Sun, Hamid Palangi, Shariq Iqbal, Ivan Vulić, Anna Korhonen, and Sercan Ö. Arik. 2025d. Multi-Agent Design: Optimizing Agents with Better Prompts and Topologies. (2025). [arXiv:2502.02533](#)
- Meng Zhou, Bei Li, Jiahao Liu, Xiaowen Shi, Yang Bai, Rongxiang Weng, Jingang Wang, and Xunliang Cai. 2025b. Libra: Assessing and Improving Reward Model by Learning to Think. (2025). [arXiv:2507.21645](#)
- Wangchunshu Zhou, Yixin Ou, Shengwei Ding, Long Li, Jialong Wu, Tiannan Wang, Jiamin Chen, Shuai Wang, Xiaohua Xu, Ningyu Zhang, Huajun Chen, and Yuchen Eleanor Jiang. 2024a. Symbolic Learning Enables Self-Evolving Agents. (2024). [arXiv:2406.18532](#)
- Wenxuan Zhou, Shujian Zhang, Lingxiao Zhao, and Tao Meng. 2024d. T-REG: Preference Optimization with Token-Level Reward Regularization. (2024). [arXiv:2412.02685](#)
- Yan Zhou and Yanguang Chen. 2025. Adaptive heterogeneous multi-agent debate for enhanced educational and factual reasoning in large language models. *Journal of King Saud University Computer and Information Sciences* 37, 10 (2025), 330.
- Yongchao Zhou, Andrei Ioan Muresanu, Ziwen Han, Keiran Paster, Silviu Pitis, Harris Chan, and Jimmy Ba. 2023. Large Language Models Are Human-Level Prompt Engineers. In *International Conference on Learning Representations*. [arXiv:2211.01910](#)
- Yuxuan Zhou, Yubin Wang, Bin Wang, Chen Ning, Xien Liu, Ji Wu, and Jianye Hao. 2025e. Enhancing the Medical Context-Awareness Ability of LLMs via Multifaceted Self-Refinement Learning. *arXiv preprint arXiv:2511.10067* (2025).
- Yilun Zhou, Austin Xu, Peifeng Wang, Caiming Xiong, and Shafiq Joty. 2025f. Evaluating Judges as Evaluators: The JETTS Benchmark of LLM-as-Judges as Test-Time Scaling Evaluators. (2025). [arXiv:2504.15253](#)
- Yifei Zhou, Andrea Zanette, Jiayi Pan, Sergey Levine, and Aviral Kumar. 2024c. ARCHER: Training Language Model Agents via Hierarchical Multi-Turn RL. (2024). [arXiv:2402.19446](#)
- Adam Zweiger, Jyothish Pari, Han Guo, Ekin Akyürek, Yoon Kim, and Pulkit Agrawal. 2025. Self-Adapting Language Models. (2025). [arXiv:2506.10943](#)